Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

Wen, Zehai

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\boldsymbol{x}_i\}_{i=1}^n$ where $\boldsymbol{x}_i \in \mathbb{R}^p$ for all $i = 1, \ldots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean $\mathbf{0}$), the probabilistic principal component analysis model is a statistical model that treats $\boldsymbol{x}_i$'s as realizations of independent random variables $\boldsymbol{X}_i = \boldsymbol{W}\boldsymbol{Z}_i + \boldsymbol{\epsilon_i}$, where $\boldsymbol{W}$ is an unknown $p \times q$ loading matrix of full rank, $\boldsymbol{Z}_i \sim N_q(\mathbf{0}, \boldsymbol{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon_i} \sim N_p(\mathbf{0}, \sigma^2\boldsymbol{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_n, \boldsymbol{\epsilon}_1, \ldots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\boldsymbol{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ. Θ is thus the collection of all tuples $(\boldsymbol{W}, \sigma^2)$ such that $\boldsymbol{W}$ is a real $p \times q$ matrix with rank$(\boldsymbol{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \left\{N_p\left(\mathbf{0}, \boldsymbol{W}_\theta\boldsymbol{W}_\theta^T + \sigma_\theta^2\boldsymbol{I}_p\right)\right\}_{\theta\in\Theta}. \tag{1.1}$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\boldsymbol{A}, s) \in \Theta$ and $(\boldsymbol{B}, t) \in \Theta$, we define $(\boldsymbol{A}, s) \sim (\boldsymbol{B}, t)$ to mean that they give rise to the same covariance matrix,namely when $\boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p = \boldsymbol{B}\boldsymbol{B}^T + t\boldsymbol{I}_p$. The equivalence classes are denoted by $[(\boldsymbol{A}, s)]$ or more briefly $[\boldsymbol{A}, s]$. Unless otherwise mentioned, $(\boldsymbol{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\boldsymbol{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\boldsymbol{A}, s] = \{(\boldsymbol{A}\boldsymbol{Q}, s) \mid \boldsymbol{Q} \in O(q)\}$. Therefore, $M := \Theta/\sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee(2013)], Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension $pq + 1 - \frac{q(q-1)}{2}$ and the map $\pi : \Theta \to M$ defined by

$$\pi((\boldsymbol{A}, s)) = [\boldsymbol{A}, s] \tag{1.2}$$

is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomertic with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee(2018)], Theorem 2.28), meaning that

$$g^\Theta_{(\boldsymbol{A},s)}((\boldsymbol{V}, c), (\boldsymbol{U}, b)) = g_{[\boldsymbol{A},s]}\left(d\pi_{(A,s)}((\boldsymbol{V}, c)), d\pi_{(A,s)}((\boldsymbol{U}, b))\right), \tag{1.3}$$

where $g^\Theta_{(\boldsymbol{A},s)}$ is the standard Euclidean metric given by

$$g^\Theta_{(\boldsymbol{A},s)}((\boldsymbol{V}, c), (\boldsymbol{U}, b)) = \operatorname{tr}\left(\boldsymbol{V}\boldsymbol{U}^T\right) + cb, \tag{1.4}$$

whenever $(\boldsymbol{A}, s) \in \Theta$, $(\boldsymbol{V}, c)$, $(\boldsymbol{U}, b) \in T_{(\boldsymbol{A},s)}\Theta \cong \mathbb{R}^{p\times q} \times \mathbb{R}$. For later convenience, let us define two functions $\Phi : \Theta \to \mathbb{R}^{p\times p}$ and $\boldsymbol{\Sigma} : M \to \mathbb{R}^{p\times p}$ by setting, for every $[\boldsymbol{A}, s] \in M$:

$$\boldsymbol{\Sigma}([\boldsymbol{A}, s]) := \Phi((\boldsymbol{A}, s)) := \boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p. \tag{1.5}$$

In other words, $\boldsymbol{\Sigma} \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\mathbf{0}, \boldsymbol{\Sigma}([\boldsymbol{A}, s]))\}_{[\boldsymbol{A},s]\in M} \tag{1.6}$$

and this is an identifiable model we shall work with.

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces.

Proposition 1.

1. The map $\boldsymbol{\Sigma}$ defined in (1.5) is a smooth injective immersion.
2. At every $(\boldsymbol{A}, s) \in \Theta$, the verticle tangent space
$$V_{(\boldsymbol{A},s)} := \ker(d\pi_{(\boldsymbol{A},s)}) = \ker(d\Phi_{(\boldsymbol{A},s)}) \tag{1.7}$$
equals to the collection of all tangent vectors of the form $(\boldsymbol{A}\boldsymbol{\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.
3. At every $(\boldsymbol{A}, s) \in \Theta$, the horizontal tangent space $H_{(\boldsymbol{A},s)} := V^{\perp}_{(\boldsymbol{A},s)}$ equals to the collection of all tangent vectors of the form $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $q \times q$ matrices $\boldsymbol{V}$ satisfying $\boldsymbol{V}^T\boldsymbol{A} = \boldsymbol{A}^T\boldsymbol{V}$.

Proof. Fix a point $(\boldsymbol{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(d\Phi_{(\boldsymbol{A},s)})$, where Φ is given in (1.5). For $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ by setting $\gamma(t) = (\boldsymbol{A} + t\boldsymbol{V}, s + tc)$. Then:

$$d\Phi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c)) = \frac{d}{dt}\,|_{t=0}\, \Phi(\gamma(t)) = \boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p.$$

If $(\boldsymbol{V}, c) \in \ker(d\Phi_{(\boldsymbol{A},s)})$, then

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p = \mathbf{0}. \tag{1.8}$$

If $\boldsymbol{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\boldsymbol{A}$, then $\boldsymbol{A}^T\boldsymbol{u} = \mathbf{0}$ and

$$0 = \boldsymbol{u}^T(\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T + c\boldsymbol{I}_p)\boldsymbol{u} = c\,\boldsymbol{u}^T\boldsymbol{u}.$$

Given that $\boldsymbol{u} \neq \mathbf{0}$, we must have $c = 0$ and (1.8) reduces to

$$\boldsymbol{A}\boldsymbol{V}^T + \boldsymbol{V}\boldsymbol{A}^T = \mathbf{0}. \tag{1.9}$$

Apply $\boldsymbol{u}$ to the right to get:

$$\boldsymbol{A}\boldsymbol{V}^T\boldsymbol{u} = \mathbf{0}.$$

Given that $\boldsymbol{A}$ has full rank, rank nullity theorem forces

$$\boldsymbol{V}^T\boldsymbol{u} = \mathbf{0}.$$

Therefore, we have proven that every vector $\boldsymbol{u}$ orthgonal to columns vectors of $\boldsymbol{A}$ will also be orthgonal to column vectors of $\boldsymbol{V}$. This means $\boldsymbol{V} = \boldsymbol{A\Omega}$ for some $q \times q$ matrix $\boldsymbol{\Omega}$. Substitute this back to (1.9) gives

$$\boldsymbol{A}(\boldsymbol{\Omega}^T + \boldsymbol{\Omega})\boldsymbol{A}^T = \boldsymbol{0}.$$

Applying $(\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T$ to the left and $\boldsymbol{A}(\boldsymbol{A}^T\boldsymbol{A})^{-1}$ to the right yields $\boldsymbol{\Omega}^T + \boldsymbol{\Omega} = \boldsymbol{0}$. Therefore, $\boldsymbol{\Omega}$ is skew-symmetric. Conversely, (1.9) is satisfied by $\boldsymbol{V} = \boldsymbol{A\Omega}$ with any skew-symmetric matrix $\boldsymbol{\Omega}$. We have proven that $\ker(d\Phi_{(\boldsymbol{A},s)})$ equals to the collection of all tangent vectors of the form $(\boldsymbol{A\Omega}, 0) \in T_{(\boldsymbol{A},s)}\Theta$, with $\boldsymbol{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.7). Clearly, $\boldsymbol{\Sigma}$ is well-defined and is injective. The smoothness of $\boldsymbol{\Sigma}$ is guaranteed by relation $\boldsymbol{\Sigma} \circ \pi = \Phi$ (see for example [Lee(2013)] Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\boldsymbol{A},s)} = d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]} \circ d\pi_{(\boldsymbol{A},s)}. \tag{1.10}$$

It follows immediately that $\ker(d\Phi_{(\boldsymbol{A},s)}) \subseteq \ker(d\pi_{(\boldsymbol{A},s)})$. The other direction will follow if we can prove that $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective, or equivalently that $\boldsymbol{\Sigma}$ is an immersion.

To do so, suppose that $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ is such that $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = 0$. We prove that $\boldsymbol{v} = \boldsymbol{0}$. Let $(\boldsymbol{V}, c) \in T_{(\boldsymbol{A},s)}\Theta$ be the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$, meaning that $(\boldsymbol{V}, c)$ is the unique horizontal tangent vector at $(\boldsymbol{A}, s)$ such that $\boldsymbol{v} = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{V}, c))$. (1.10) then implies that $(\boldsymbol{V}, c) \in \ker(d\Phi_{(\boldsymbol{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\boldsymbol{V} = \boldsymbol{A\Omega}$ for some skew-symmetric $\boldsymbol{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \to \Theta$ defined by $\gamma(t) = (\boldsymbol{A}e^{t\boldsymbol{\Omega}}, s)$. Clearly $\gamma(0) = (\boldsymbol{A}, s)$ and $\gamma'(0) = (\boldsymbol{A\Omega}, 0)$. Moreover, the matrix $e^{t\boldsymbol{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\boldsymbol{\Omega}$ so that $\pi(\gamma(t)) = [\boldsymbol{A}, s]$ for all t. We have

$$\boldsymbol{v} = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{A\Omega}, 0)) = d\pi_{(\boldsymbol{A},s)}(\gamma'(0)) = \frac{d}{dt}\,|_{t=0}\ \pi(\gamma(t)) = \boldsymbol{0} \in T_{[\boldsymbol{A},s]}M.$$

Therefore, $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is injective and the verticle tangent space is the same as $\ker(d\Phi_{(\boldsymbol{A},s)})$.

To compute the horizontal tangent space, any $(\boldsymbol{V}, c) \in V^{\perp}_{(\boldsymbol{A},s)}$ will satisfy

$$0 = g_{(\boldsymbol{A},s)}\left((\boldsymbol{V}, c), (\boldsymbol{A\Omega}, 0)\right) = \operatorname{tr}\left(\boldsymbol{A}^T\boldsymbol{V\Omega}\right) \tag{1.11}$$

for all skew-symmetric $\boldsymbol{\Omega}$. Decomposing $\boldsymbol{A}^T\boldsymbol{V} = \boldsymbol{S} + \boldsymbol{K}$, where $\boldsymbol{S}$ is the symmetric part and $\boldsymbol{K}$ is the skew-symmetric part, we find out that

$$0 = \operatorname{tr}(\boldsymbol{S\Omega}) + \operatorname{tr}(\boldsymbol{K\Omega}) = 0 + \operatorname{tr}(\boldsymbol{K\Omega}) = \operatorname{tr}(\boldsymbol{K\Omega}).$$

Setting $\boldsymbol{\Omega} = \boldsymbol{K}^T$ will yield that $\boldsymbol{K} = \boldsymbol{0}$. Therefore, $\boldsymbol{A}^T\boldsymbol{V}$ is a symmetric matrix. Conversely, any $\boldsymbol{V}$ such that $\boldsymbol{A}^T\boldsymbol{V}$ is symmetric will satisfy (1.11). The proof is complete. □

Proposition 2. Given $[\boldsymbol{A}, s] \in M$ and $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$, geodesics in M starting at $[\boldsymbol{A}, s]$ with initial velocity $\boldsymbol{v}$ takes the format of curves $\gamma : I \to M$, where I is an open interval containing 0, defined by $\gamma(t) = [\boldsymbol{A} + t\boldsymbol{V}, s + tc]$, where $(\boldsymbol{V}, c) \in H_{(\boldsymbol{A},s)}$ is the horizontal lift of $\boldsymbol{v}$ at $(\boldsymbol{A}, s)$. Consequently, the exponential map is computed as $\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = [\boldsymbol{A} + \boldsymbol{V}, s + c]$, well-defined over all $\boldsymbol{v}$ such that $\sqrt{g_{[\boldsymbol{A},s]}(\boldsymbol{v}, \boldsymbol{v})} = |\boldsymbol{v}| < \min(\sigma_{\min}(\boldsymbol{A}), s)$, where $\sigma_{\min}(\boldsymbol{A})$ is the smallest singular value of $\boldsymbol{A}$.

Proof. First of all, let us quickly check that the curve γ is well defined. Indeed, if $(\boldsymbol{AQ}, s)$, where $\boldsymbol{Q} \in O(q)$, is another representative of $[\boldsymbol{A}, s]$, let $(\boldsymbol{V}', c')$ be the horizontal lift of $\boldsymbol{v}$ to $(\boldsymbol{AQ}, s)$. With the function $R_{\boldsymbol{Q}} : \Theta \to \Theta$ defined by $R_{\boldsymbol{Q}}((\boldsymbol{A}, s)) = (\boldsymbol{AQ}, s)$, we have

$$\boldsymbol{v} = d\pi_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = d(\pi \circ R_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = d\pi_{(\boldsymbol{AQ},s)}(dR_{\boldsymbol{Q}})_{(\boldsymbol{A},s)}(\boldsymbol{V}, c) = d\pi_{(\boldsymbol{AQ},s)}(\boldsymbol{VQ}, c).$$

Thus, $\boldsymbol{V}' = \boldsymbol{VQ}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve.

A well-known property of geodesics in case of Riemannian submersion (see for example [O'Neill(1967)] Section 2 Corollary 2 or [Valencia(2021)] Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M. Therefore, $\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}) = [\boldsymbol{A} + \boldsymbol{V}, s + c]$ is well defined for all vectors $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ of small enough norm making sure that $(\boldsymbol{A} + \boldsymbol{V}, s + c) \in \Theta$. To have a quantified upper bound of the norm, we cite Weyl's perturbation inequality (Chapter 7.3 Problem 16 equation (7.3.15) of [Horn and Johnson(2013)]) to get

$$\sigma_{\text{min}}(\boldsymbol{A} + \boldsymbol{V}) \geqslant \sigma_{\text{min}}(\boldsymbol{A}) - \sigma_{\text{max}}(\boldsymbol{V}) \geqslant \sigma_{\text{min}}(\boldsymbol{A}) - |\boldsymbol{v}| > 0,$$

whenever $|\boldsymbol{v}| < \min(\sigma_{\text{min}}(\boldsymbol{A}), s)$. Here $\sigma_{\text{max}}(\boldsymbol{V})$ denotes the largest singular value of $\boldsymbol{V}$, who is smaller than the Euclidean norm of $\boldsymbol{V}$ and hence norm of $\boldsymbol{v}$. Thus, for these $\boldsymbol{v}$, $\mathrm{rank}(\boldsymbol{A} + \boldsymbol{V}) = q$ and the exponential map is well-defined. □

To compute the logarithmic map, we need first preparatory lemmas that work out an orthogonal matrix based on given points in M.

Lemma 3. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix and $\boldsymbol{Q} \in O(q)$ are such that $\boldsymbol{LQ}$ is symmetric, then there exists matrices $\boldsymbol{R} \in O(q)$, $\boldsymbol{P} \in O(q)$, a $q \times q$ diagonal matrix $\boldsymbol{D}$ whose diagonal entries are 1 or -1, and a $q \times q$ diagonal matrix $\boldsymbol{S}$ containing the singular values of $\boldsymbol{L}$ in increasing order such that $\boldsymbol{L} = \boldsymbol{RSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDR}^T$.

Proof. Let us start by applying the singular value decomposition of $\boldsymbol{L}$ to get matrices $\boldsymbol{R}' \in O(q)$, $\boldsymbol{P}' \in O(q)$ and $\boldsymbol{S}$ such that $\boldsymbol{L} = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T$. Substituting the given condition that $\boldsymbol{LQ}$ is symmetric will yield that $\boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$, where $\boldsymbol{J} := (\boldsymbol{P}')^T\boldsymbol{Q}\boldsymbol{R}'$.

Clearly $\boldsymbol{J} \in O(q)$. The equation

$$\boldsymbol{S}^2 = \boldsymbol{SJJ}^T\boldsymbol{S} = (\boldsymbol{SJ})(\boldsymbol{SJ})^T = (\boldsymbol{J}^T\boldsymbol{S})(\boldsymbol{J}^T\boldsymbol{S})^T = \boldsymbol{J}^T\boldsymbol{S}^2\boldsymbol{J}$$

shows that

$$\boldsymbol{JS}^2 = \boldsymbol{S}^2\boldsymbol{J}.$$

Looking at each entry of both size, we end up with

$$\boldsymbol{J}_{ij}(\boldsymbol{S}_{jj}^2 - \boldsymbol{S}_{ii}^2) = 0$$

for $i, j = 1, \ldots q$. Given that $\boldsymbol{L}$ is invertible, its singular values are all strictly positive. Therefore, in fact we have

$$\boldsymbol{J}_{ij}(\boldsymbol{S}_{jj} - \boldsymbol{S}_{ii}) = 0 \tag{1.12}$$

for all i and all j so that $\boldsymbol{JS} = \boldsymbol{SJ} = \boldsymbol{J}^T\boldsymbol{S}$. Invert $\boldsymbol{S}$ yields that $\boldsymbol{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\boldsymbol{J}^2 = \boldsymbol{I}_q$ and the eigenvalues of $\boldsymbol{J}$ must be 1 or -1. Another look at (1.12) shows that $\boldsymbol{J}$ is a block-diagonal matrix, diagonal if and only if all diagonal entries of $\boldsymbol{S}$ are distinct. Furthermore, each block of $\boldsymbol{J}$ must still be orthogonal and symmetric.

We now diagonalize $\boldsymbol{J}$ based on the above information. Suppose that $\boldsymbol{S} = \oplus_{i=1}^k \boldsymbol{s}_i \boldsymbol{I}_{m_i}$, where $\boldsymbol{s}_1 > \ldots > \boldsymbol{s}_k$ are the unique diagonal entries of $\boldsymbol{S}$ and each repeats for m_i times. Thus, $\boldsymbol{J}$ has k blocks. Write the ith block by $\boldsymbol{J}_i$ so that $\boldsymbol{J} = \oplus_{i=1}^k \boldsymbol{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\boldsymbol{J}_i = \boldsymbol{K}_i\boldsymbol{D}_i\boldsymbol{K}_i^T$, where $\boldsymbol{K}_i \in O(m_i)$ and $\boldsymbol{D}_i$ is an $m_i \times m_i$ diagonal matrix with 1 or -1 diagonal entry. Put $\boldsymbol{K} = \oplus_{i=1}^k \boldsymbol{K}_i$ and $\boldsymbol{D} = \oplus_{i=1}^k \boldsymbol{D}_i$. Then $\boldsymbol{K} \in O(q)$, $\boldsymbol{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\boldsymbol{J} = \boldsymbol{KDK}^T$. This involved construction leads us to the outcome that $\boldsymbol{K}$ in fact commutes with $\boldsymbol{S}$:

$$\boldsymbol{KS} = \bigoplus_{i=1}^k \boldsymbol{s}_i\boldsymbol{K}_i = \boldsymbol{SK}.$$

Put $\boldsymbol{P} = \boldsymbol{P}'\boldsymbol{K}$, $\boldsymbol{R} = \boldsymbol{R}'\boldsymbol{K}$. We have

$$\boldsymbol{Q} = \boldsymbol{P}'\boldsymbol{J}(\boldsymbol{R}')^T = \boldsymbol{P}'\boldsymbol{KDK}^T(\boldsymbol{R}')^T = \boldsymbol{PDR}^T$$

and

$$\boldsymbol{RSP}^T = \boldsymbol{R}'\boldsymbol{KSK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{SKK}^T(\boldsymbol{P}')^T = \boldsymbol{R}'\boldsymbol{S}(\boldsymbol{P}')^T = \boldsymbol{L}.$$

This completes the proof. □

Lemma 4. If $\boldsymbol{L}$ is an invertible $q \times q$ matrix with singular value decomposition $\boldsymbol{L} = \boldsymbol{RSP}^T$, then

$$\boldsymbol{PR}^T = (\boldsymbol{L}^T\boldsymbol{L})^{-\frac{1}{2}}\boldsymbol{L}^T.$$

This means, although neither $\boldsymbol{P}$ nor $\boldsymbol{R}$ is uniquely determined, $\boldsymbol{PR}^T$ is uniquely determined.

Proof. Given that

$$\boldsymbol{L}^T\boldsymbol{L} = \boldsymbol{PS}^2\boldsymbol{P}^T,$$

we have

$$(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}} = \boldsymbol{PSP}^T.$$

Therefore:

$$\boldsymbol{L} = \boldsymbol{RP}^T\boldsymbol{PSP}^T = \boldsymbol{RP}^T(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}.$$

The result follows from inverting $(\boldsymbol{L}^T\boldsymbol{L})^{\frac{1}{2}}$ and take transpose. □

Proposition 5. For every $[\boldsymbol{A}, s] \in M$ there exists an open neighborhood $U \subset M$ containing $[\boldsymbol{A}, s]$ such that

1. For every $[\boldsymbol{B}, t] \in U$, $\boldsymbol{A}^T\boldsymbol{B}$ is invertible.
2. The logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]} : U \to T_{[\boldsymbol{A},s]}M$ is well-defined and is smooth in U. In fact:

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{BQ}(\boldsymbol{A}, \boldsymbol{B}) - \boldsymbol{A}, t - s)),$$

where

$$\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B}) = (\boldsymbol{B}^T\boldsymbol{AA}^T\boldsymbol{B})^{-\frac{1}{2}}\boldsymbol{B}^T\boldsymbol{A} \in O(q).$$

Proof. First of all, if $\boldsymbol{R} \in O(q)$ and $\boldsymbol{P} \in O(q)$, then $\boldsymbol{Q}(\boldsymbol{AP}, \boldsymbol{BR}) = \boldsymbol{R}^T\boldsymbol{Q}(\boldsymbol{A}, \boldsymbol{B})\boldsymbol{P}$ and the same computation as in the proof of Proposition 2 shows that the formula of logarithmic map is indeed well-defined.

Given that the determinant is a continuous function and $\det\big(\boldsymbol{A}^T\boldsymbol{A}\big) > 0$, there exists a neighborhood $U_\Theta \subset \Theta$ containing $(\boldsymbol{A}, s)$ such that $\det\big(\boldsymbol{A}^T\boldsymbol{B}\big) > 0$ whenever $(\boldsymbol{B}, t) \in U_\Theta$. Thus, for any $[\boldsymbol{B}, t] \in U_1 := \pi(U_\Theta) \subset M$, $\boldsymbol{A}^T\boldsymbol{B}$ is invertible.

Next, it is a well-known fact that $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is a local diffeomorphism and thus there exists an open neighborhood $U_2 \subset M$ containing $[\boldsymbol{A}, s]$ such that the logarithmic map $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}$ is well-defined and is smooth in U_2.

We now compute $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}$ for $[\boldsymbol{B}, t] \in U := U_1 \cap U_2$. Suppose that $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ satisfies $\boldsymbol{v} = \mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t])$. By Proposition 2, $[\boldsymbol{B}, t] = [\boldsymbol{A} + \boldsymbol{V}, s + c]$, where $(\boldsymbol{V}, c)$ is the horizontal lift of $\boldsymbol{v}$ to $(\boldsymbol{A}, s)$. It follows that $c = t - s$ and that there exists a $\boldsymbol{Q} \in O(q)$ such that $\boldsymbol{V} = \boldsymbol{BQ} - A$. Or equivalently:

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{BQ} - \boldsymbol{A}, t - s)). \tag{1.13}$$

Given that $(\boldsymbol{V}, c)$ is a horizontal tangent vector, Proposition 1 shows that

$$\boldsymbol{A}^T(\boldsymbol{BQ} - \boldsymbol{A}) = (\boldsymbol{BQ} - \boldsymbol{A})^T\boldsymbol{A}$$

and consequently $\boldsymbol{A}^T\boldsymbol{BQ}$ is symmetric. Lemma 3 with $\boldsymbol{L} = \boldsymbol{A}^T\boldsymbol{B}$ shows the existence of $\boldsymbol{R}$, $\boldsymbol{P} \in O(q)$, a diagonal matrix $\boldsymbol{S}$ with the singular values of $\boldsymbol{A}^T\boldsymbol{B}$ in increasing order, and a $q \times q$ diagonal matrix $\boldsymbol{D}$ with diagonal entries 1 or -1 such that $\boldsymbol{A}^T\boldsymbol{B} = \boldsymbol{RSP}^T$ and $\boldsymbol{Q} = \boldsymbol{PDR}^T$. (1.13) now becomes

$$\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B}, t]) = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{BPDR}^T - \boldsymbol{A}, t - s)). \tag{1.14}$$

Now let $\boldsymbol{B} \to \boldsymbol{A}$ and $t \to s$. The left hand size of (1.14) now becomes $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{A},s]) = \boldsymbol{0} \in T_{[\boldsymbol{A},s]}M$. The singular value decomposition converges to the orthogonal diagonalization because $\boldsymbol{A}^T\boldsymbol{A}$ is symmetric, meaning that $\boldsymbol{P} = \boldsymbol{R}$. Thus

$$\boldsymbol{A}\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T - \boldsymbol{A} = \boldsymbol{0}.$$

Therefore, $\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^T = \boldsymbol{I}_q$ and $\boldsymbol{D} = \boldsymbol{I}_q$. Lemma 4 then shows that

$$\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{R}^T = (\boldsymbol{B}^T\boldsymbol{A}\boldsymbol{A}^T\boldsymbol{B})^{-\frac{1}{2}}\boldsymbol{B}^T\boldsymbol{A}.$$

This completes the proof. □

Remark 6. Unfortunately, the logarithmic map is not globally defined in the standard way even over all $[\boldsymbol{B},t]$ such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible. For a counterexample, take $p=2$, $q=1$, $\boldsymbol{A} = [1,0]^T$, $\boldsymbol{B} = [1,1]^T$ and $s=t=1$. In this case, $O(1) = \{1,-1\}$. Then, although $\boldsymbol{A}^T\boldsymbol{B} = 1$ is invertible, we still have

$$\operatorname{Exp}_{[\boldsymbol{A},s]}\left(d\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}-\boldsymbol{A},0))\right) = [\boldsymbol{B},1] = [-\boldsymbol{B},1] = \operatorname{Exp}_{[\boldsymbol{A},s]}\left(d\pi_{(\boldsymbol{A},s)}((-\boldsymbol{B}-\boldsymbol{A},0))\right).$$

The reason is that, even when we assume $\boldsymbol{A}^T\boldsymbol{B}$ is invertible, deducing $\boldsymbol{D} = \boldsymbol{I}_q$ relies on the local smoothness of the logarithmic map and the procedure of letting $[\boldsymbol{B},t] \to [\boldsymbol{A},s]$. If $[\boldsymbol{B},t]$ is far away from $[\boldsymbol{A},s]$, then $\boldsymbol{Q} = \boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T$ with some $\boldsymbol{D} \neq \boldsymbol{I}_q$ can still produce a valid horizontal vector whose image under $\operatorname{Exp}_{[\boldsymbol{A},s]}$ is $[\boldsymbol{B},t]$, so that $\operatorname{Exp}_{[\boldsymbol{A},s]}$ fails to be injective and admits no globally defined inverse.

Nevertheless, the choice $\boldsymbol{D} = \boldsymbol{I}_q$ has an interesting property of selecting the geodesic of minimal speed and length when $[\boldsymbol{B},t]$ is any point such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible. To see this, consider a general candidate $\boldsymbol{Q}' = \boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T$ and write $\boldsymbol{v}' = d\pi_{(\boldsymbol{A},s)}((\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s))$, so that $(\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s)$ is the horizontal lift of $\boldsymbol{v}'$ at $(\boldsymbol{A},s)$ and $\operatorname{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}') = [\boldsymbol{B},t]$. Let us compute the squared norm of $\boldsymbol{v}'$:

$$\begin{aligned} g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}') &= g^{\ominus}_{(\boldsymbol{A},s)}\left((\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s),(\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A},t-s)\right) \\ &= \operatorname{tr}\left((\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A})(\boldsymbol{B}\boldsymbol{Q}'-\boldsymbol{A})^T\right) + (t-s)^2 = \operatorname{tr}\big(\boldsymbol{B}\boldsymbol{B}^T\big) + \operatorname{tr}\big(\boldsymbol{A}\boldsymbol{A}^T\big) - 2\operatorname{tr}(\boldsymbol{S}\boldsymbol{D}) + (t-s)^2, \end{aligned}$$

where the last equality substituted $\boldsymbol{A}^T\boldsymbol{B}\boldsymbol{Q}' = \boldsymbol{R}\boldsymbol{S}\boldsymbol{P}^T\boldsymbol{P}\boldsymbol{D}\boldsymbol{R}^T = \boldsymbol{R}\boldsymbol{S}\boldsymbol{D}\boldsymbol{R}^T$ and used the cyclic property of the trace. The fact that the diagonal entries of $\boldsymbol{S}$ are strictly positive show that $g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}')$ is minimized precisely when $\boldsymbol{D} = \boldsymbol{I}_q$. Given that the geodesic starting at $[\boldsymbol{A},s]$ with initial velocity $\boldsymbol{v}'$ reaches $[\boldsymbol{B},t]$ at time 1 with constant speed $\sqrt{g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}')}$, its length is $\sqrt{g_{[\boldsymbol{A},s]}(\boldsymbol{v}',\boldsymbol{v}')}$. The choice $\boldsymbol{D} = \boldsymbol{I}_q$ gives the slowest and hence shortest geodesic. To illustrate, let us go back to the counterexample in the previous paragraph. Put $\boldsymbol{v}_\pm := d\pi_{(\boldsymbol{A},s)}((\pm\boldsymbol{B}-\boldsymbol{A},0))$. Direct computation shows $g_{[\boldsymbol{A},s]}(\boldsymbol{v}_+,\boldsymbol{v}_+) = 1$ and $g_{[\boldsymbol{A},s]}(\boldsymbol{v}_-,\boldsymbol{v}_-) = 5$ so that the option corresponding to $\boldsymbol{D} = \boldsymbol{I}_1 = 1$ gives the slowest speed. Therefore, it is actually possible to define $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}$ over all $[\boldsymbol{B},t]$ such that $\boldsymbol{A}^T\boldsymbol{B}$ is invertible by manually asking it to output the minimal speed candidate and this definition will yield the same $\boldsymbol{Q}(\boldsymbol{A},\boldsymbol{B})$ formula.

However, the situation totally breaks down in case $\boldsymbol{A}^T\boldsymbol{B}$ is not invertible. Consider again $p=2$, $q=1$, $\boldsymbol{A} = [1,0]^T$, $s=t=1$, but this time $\boldsymbol{B} = [0,1]^T$. Then $g_{[\boldsymbol{A},s]}(\boldsymbol{v}_+,\boldsymbol{v}_+) = g_{[\boldsymbol{A},s]}(\boldsymbol{v}_-,\boldsymbol{v}_-) = 2$ and there are two distinct geodesics of the same speed so $\operatorname{Exp}^{-1}_{[\boldsymbol{A},s]}([\boldsymbol{B},t])$ cannot be well-defined.

(Picture: there is a hole at $[0,0]^T$. Starting $\boldsymbol{A} = [1,0]^T$, as $\boldsymbol{B} = [0,1]^T$ is glued to $[0,-1]^2$, $\boldsymbol{v}_+$ points from $[1,0]^T$ to $[0,1]^T$, $\boldsymbol{v}_-$ points from $[1,0]^T$ to $[0,-1]^T$)

Given the above situation and (the expectation that) our work should not require a global definition, we stick to the standard definition of logarithmic map only defined locally around $[\boldsymbol{A},s]$.

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.6) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in [Sun et al.(2026)Sun, Lin, and Liu], definition III.1 and III.2.

Let us introduce a few notations. For each $[\boldsymbol{A}, s] \in M$, let

$$p(\boldsymbol{x}; [\boldsymbol{A}, s]) = (2\pi)^{-\frac{p}{2}} \det{}^{-\frac{1}{2}}(\boldsymbol{\Sigma}([\boldsymbol{A}, s])) \exp\left(-\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{\Sigma}([\boldsymbol{A}, s]))^{-1}\boldsymbol{x}\right) \tag{1.15}$$

be the density of $N_p(\boldsymbol{0}, \boldsymbol{\Sigma}([\boldsymbol{A}, s]))$. Define two maps $l_{[\boldsymbol{A},s]}$ and $r_{[\boldsymbol{A},s]}$ with domain $T_{[\boldsymbol{A},s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$ and codomain $L^2(\mathbb{R}^p)$ by setting, for all $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$,

$$l_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = \log\left(p(\boldsymbol{x}; \mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))\right) \tag{1.16}$$

and

$$r_{[\boldsymbol{A},s]}(\boldsymbol{v})(\boldsymbol{x}) = \sqrt{p(\boldsymbol{x}; \mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v}))}. \tag{1.17}$$

To show that $\mathcal{P}$ is differentiable in quadratic mean, we must, by definition, show that for all $[\boldsymbol{A}, s] \in M$ that $r_{[\boldsymbol{A},s]}$ is Fréchet differentiable at $\boldsymbol{v} = \boldsymbol{0}$ and give an explicit formula for its Fréchet derivative. We start by the computation of its Gâteaux derivative.

Lemma 7. The Gâteaux derivative of $r_{[\boldsymbol{A},s]}$ at $\boldsymbol{v} \in T_{[\boldsymbol{A},s]}M$ in the direction of $\boldsymbol{h} \in T_{[\boldsymbol{A},s]}M$ is

Proof. Given two vectors $\boldsymbol{v}$ and $\boldsymbol{h}$ of small norm, suppose that they respectively lift to $(\boldsymbol{V}, c)$ and $(\boldsymbol{U}, b)$ at $(\boldsymbol{A}, s)$. Then, for any $t \in \mathbb{R}$, $\boldsymbol{v} + t\boldsymbol{h}$ lifts to $(\boldsymbol{V} + t\boldsymbol{U}, c + tb)$ at $(\boldsymbol{A}, s)$ and

$$\boldsymbol{\Sigma}(\mathrm{Exp}_{[\boldsymbol{A},s]}(\boldsymbol{v} + t\boldsymbol{h})) = \boldsymbol{\Sigma}([\boldsymbol{A} + \boldsymbol{V} + t\boldsymbol{U}, s + c + tb]) = (\boldsymbol{A} + \boldsymbol{V} + t\boldsymbol{U})(\boldsymbol{A} + \boldsymbol{V} + t\boldsymbol{U})^T + (s + c + tb)\boldsymbol{I}_p$$

$\square$

Chapter 2

Draft

We now investigate some properties of the model when the parameter space is now M. The goal of this section is to show that the model, viewed as a family of probability measures indexed by the quotient manifold M, is *differentiable in quadratic mean*, and to compute its score operator and its Fisher information operator in closed form. This is the entry point of the Le Cam theory on parameter manifolds developed in [Sun et al.(2026)Sun, Lin, and Liu] and is what eventually licences the asymptotic normality statement we are after. Throughout we place ourselves in the framework of that paper with their parameter manifold taken to be our M and their parameter of interest ψ taken to be the identity, so that no distinction between the parameter and the parameter of interest needs to be carried along.

Let us first fix some notation. We write $\mathrm{Sym}(p)$ for the space of real symmetric $p\times p$ matrices and $\mathrm{Sym}_{++}(p) \subset \mathrm{Sym}(p)$ for the open convex cone of symmetric positive definite ones. For a real matrix $\boldsymbol{B}$ we write $\|\boldsymbol{B}\|_F$ for its Frobenius norm, $\|\boldsymbol{B}\|_2$ for its spectral norm, and $\sigma_{\min}(\boldsymbol{B})$ for its smallest singular value. For $v \in T_{[\boldsymbol{A},s]}M$ we write $\|v\| := \sqrt{g_{[\boldsymbol{A},s]}(v,v)}$ and $\bar{v} \in H_{(\boldsymbol{A},s)}$ for the horizontal lift of v at $(\boldsymbol{A}, s)$, so that (1.3) says precisely that $v \mapsto \bar{v}$ is a linear isometry from $T_{[\boldsymbol{A},s]}M$ onto $H_{(\boldsymbol{A},s)}$.

The statistical model on M

For $(\boldsymbol{A}, s) \in \Theta$ put

$$\boldsymbol{\Sigma}(\boldsymbol{A}, s) := \boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p, \tag{2.1}$$

which lies in $\mathrm{Sym}_{++}(p)$ because $\boldsymbol{A}\boldsymbol{A}^T \succeq \boldsymbol{0}$ and $s > 0$. By the very definition of the equivalence relation $\sim$, the map (2.1) is constant on equivalence classes and separates distinct ones. It therefore descends to an *injective* map on M, which we still denote by $\boldsymbol{\Sigma}$ and evaluate as $\boldsymbol{\Sigma}[\boldsymbol{A}, s] := \boldsymbol{\Sigma}(\boldsymbol{A}, s)$. Consequently the family

$$\mathcal{P} := \left\{P_{[\boldsymbol{A},s]} \mid [\boldsymbol{A}, s] \in M\right\}, \qquad \frac{dP_{[\boldsymbol{A},s]}}{d\boldsymbol{x}}(\boldsymbol{x}) = f(\boldsymbol{x}; [\boldsymbol{A}, s]) := \frac{\exp\left(-\frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}[\boldsymbol{A}, s]^{-1}\boldsymbol{x}\right)}{(2\pi)^{p/2}\det\left(\boldsymbol{\Sigma}[\boldsymbol{A}, s]\right)^{1/2}}, \tag{2.2}$$

is well defined — the density does not depend on the representative chosen — and the model is *identifiable*, in the sense that $[\boldsymbol{A}, s] \mapsto P_{[\boldsymbol{A},s]}$ is injective. Identifiability is exactly what was lost on Θ and is recovered on M by construction; this is the whole point of passing to the quotient. Following the Le Cam theory we work with the square root of the density,

$$r(\boldsymbol{x}; [\boldsymbol{A}, s]) := \sqrt{f(\boldsymbol{x}; [\boldsymbol{A}, s])}, \tag{2.3}$$

which is a unit vector of the Hilbert space $L^2(\mathbb{R}^p, d\boldsymbol{x})$ because $\int_{\mathbb{R}^p} r(\boldsymbol{x}; [\boldsymbol{A}, s])^2\, d\boldsymbol{x} = 1$.

The next lemma records the two structural facts that make everything in this section work: the exponential map of M is *affine* in the horizontal lift (this is the content of Proposition 2), and the covariance $\boldsymbol{\Sigma}$ read through that chart is an *exact* quadratic polynomial with no remainder.

Lemma 8. Fix $(\boldsymbol{A}, s) \in \Theta$ and put $\rho(\boldsymbol{A}, s) := \min\{\sigma_{\min}(\boldsymbol{A}), s\} > 0$. Let $h \in T_{[\boldsymbol{A},s]}M$ with $\|h\| < \rho(\boldsymbol{A}, s)$ and write its horizontal lift as $\bar{h} = (\boldsymbol{V}_h, c_h) \in H_{(\boldsymbol{A},s)}$. Then:

1. $(\boldsymbol{A} + \boldsymbol{V}_h, s + c_h) \in \Theta$, so that $\operatorname{Exp}_{[\boldsymbol{A},s]}(h) = [\boldsymbol{A} + \boldsymbol{V}_h, s + c_h]$ is defined, and $h \mapsto \bar{h}$ is linear.

2. Writing $\boldsymbol{\Sigma}_0 := \boldsymbol{\Sigma}(\boldsymbol{A}, s)$ and $\boldsymbol{\Sigma}(h) := \boldsymbol{\Sigma}\left[\operatorname{Exp}_{[\boldsymbol{A},s]}(h)\right]$, we have the exact identity

$$\boldsymbol{\Sigma}(h) = \boldsymbol{\Sigma}_0 + \dot{\boldsymbol{\Sigma}}[\bar{h}] + \boldsymbol{V}_h\boldsymbol{V}_h^T, \qquad \text{where} \qquad \dot{\boldsymbol{\Sigma}}\left[(\boldsymbol{V}, c)\right] := \boldsymbol{V}\boldsymbol{A}^T + \boldsymbol{A}\boldsymbol{V}^T + c\boldsymbol{I}_p. \tag{2.4}$$

   The map $\bar{h} \mapsto \dot{\boldsymbol{\Sigma}}[\bar{h}]$ is linear with values in $\operatorname{Sym}(p)$ and is the differential at $(\boldsymbol{A}, s)$ of the map (2.1).

3. $\boldsymbol{\Sigma}(h) \succeq (s - \|h\|)\boldsymbol{I}_p \succ \boldsymbol{0}$; in particular $\boldsymbol{\Sigma}(h) \in \operatorname{Sym}_{++}(p)$.

4. The composite map $h \mapsto \dot{\boldsymbol{\Sigma}}[\bar{h}]$ does not depend on the representative of $[\boldsymbol{A}, s]$ used to define it.

Proof. (1) Since $v \mapsto \bar{v}$ is a linear isometry, it is linear and $\|\boldsymbol{V}_h\|_F^2 + c_h^2 = \left\|\bar{h}\right\|^2 = \|h\|^2$, so that $\|\boldsymbol{V}_h\|_2 \leq \|\boldsymbol{V}_h\|_F \leq \|h\|$ and $|c_h| \leq \|h\|$; the first of these comparisons holds because $\|\boldsymbol{B}\|_2^2$ is the largest eigenvalue of $\boldsymbol{B}^T\boldsymbol{B}$ while $\|\boldsymbol{B}\|_F^2 = \operatorname{tr}\left(\boldsymbol{B}^T\boldsymbol{B}\right)$ is the sum of all of them, and all are nonnegative.

We next isolate the perturbation bound being used, since the precise reason the *spectral* norm of $\boldsymbol{V}$ appears in it is worth making explicit. We claim that for any $p \times q$ real matrices $\boldsymbol{A}$, $\boldsymbol{V}$ with $q \leq p$,

$$\sigma_{\min}(\boldsymbol{A} + \boldsymbol{V}) \geq \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\|_2. \tag{2.5}$$

Because $q \leq p$, the smallest singular value has the variational description

$$\sigma_{\min}(\boldsymbol{B}) = \min_{\boldsymbol{y}\in\mathbb{R}^q,\ \|\boldsymbol{y}\|_2=1} \|\boldsymbol{B}\boldsymbol{y}\|_2 : \tag{2.6}$$

indeed $\|\boldsymbol{B}\boldsymbol{y}\|_2^2 = \boldsymbol{y}^T\boldsymbol{B}^T\boldsymbol{B}\boldsymbol{y}$, whose minimum over unit vectors is the smallest eigenvalue of the positive semidefinite matrix $\boldsymbol{B}^T\boldsymbol{B}$, and $\sigma_{\min}(\boldsymbol{B})$ is by definition the square root of that eigenvalue. Now let $\boldsymbol{y} \in \mathbb{R}^q$ be any unit vector. The triangle inequality in $\mathbb{R}^p$, followed by the fact that $\|\cdot\|_2$ is precisely the operator norm subordinate to the Euclidean vector norm, so that $\|\boldsymbol{V}\boldsymbol{y}\|_2 \leq \|\boldsymbol{V}\|_2\|\boldsymbol{y}\|_2 = \|\boldsymbol{V}\|_2$, gives

$$\|(\boldsymbol{A} + \boldsymbol{V})\boldsymbol{y}\|_2 \geq \|\boldsymbol{A}\boldsymbol{y}\|_2 - \|\boldsymbol{V}\boldsymbol{y}\|_2 \geq \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}\|_2,$$

where the second step used (2.6) for $\boldsymbol{A}$. Taking the minimum over such $\boldsymbol{y}$ and applying (2.6) to $\boldsymbol{A} + \boldsymbol{V}$ on the left proves (2.5). This is exactly how $\|\boldsymbol{V}\|_2$, rather than some other norm of $\boldsymbol{V}$, enters: it is by definition the largest amount by which $\boldsymbol{V}$ can lengthen a unit vector, which is the only feature of $\boldsymbol{V}$ the above estimate needs.

Applying (2.5) with $\boldsymbol{V} = \boldsymbol{V}_h$,

$$\sigma_{\min}(\boldsymbol{A} + \boldsymbol{V}_h) \geq \sigma_{\min}(\boldsymbol{A}) - \|\boldsymbol{V}_h\|_2 \geq \rho(\boldsymbol{A}, s) - \|h\| > 0,$$

so $\boldsymbol{A} + \boldsymbol{V}_h$ has rank q; and $s + c_h \geq s - \|h\| \geq \rho(\boldsymbol{A}, s) - \|h\| > 0$. Hence $(\boldsymbol{A} + \boldsymbol{V}_h, s + c_h) \in \Theta$ and Proposition 2 applies.

(2) Expanding,

$$(\boldsymbol{A} + \boldsymbol{V}_h)(\boldsymbol{A} + \boldsymbol{V}_h)^T + (s + c_h)\boldsymbol{I}_p = \underbrace{\boldsymbol{A}\boldsymbol{A}^T + s\boldsymbol{I}_p}_{\boldsymbol{\Sigma}_0} + \underbrace{\boldsymbol{V}_h\boldsymbol{A}^T + \boldsymbol{A}\boldsymbol{V}_h^T + c_h\boldsymbol{I}_p}_{\dot{\boldsymbol{\Sigma}}[\bar{h}]} + \boldsymbol{V}_h\boldsymbol{V}_h^T,$$

which is (2.4); there is no higher order term because the expression is quadratic to begin with. Linearity and symmetry of $\dot{\boldsymbol{\Sigma}}$ are clear, and differentiating $t \mapsto \boldsymbol{\Sigma}(\boldsymbol{A} + t\boldsymbol{V}, s + tc)$ at $t = 0$ returns $\dot{\boldsymbol{\Sigma}}\left[(\boldsymbol{V}, c)\right]$.

(3) $\boldsymbol{\Sigma}(h) - (s + c_h)\boldsymbol{I}_p = (\boldsymbol{A} + \boldsymbol{V}_h)(\boldsymbol{A} + \boldsymbol{V}_h)^T \succeq \boldsymbol{0}$ and $s + c_h \geq s - \|h\| > 0$ by (1).

(4) Let $(\boldsymbol{AQ}, s)$, $\boldsymbol{Q} \in O(q)$, be another representative. The computation in the proof of Proposition 2 shows that the horizontal lift of h at $(\boldsymbol{AQ}, s)$ is $(\boldsymbol{V}_h\boldsymbol{Q}, c_h)$. Therefore the value of (2.4) computed at that representative is

$$\boldsymbol{V}_h\boldsymbol{Q}(\boldsymbol{AQ})^T + \boldsymbol{AQ}(\boldsymbol{V}_h\boldsymbol{Q})^T + c_h\boldsymbol{I}_p = \boldsymbol{V}_h\boldsymbol{Q}\boldsymbol{Q}^T\boldsymbol{A}^T + \boldsymbol{A}\boldsymbol{Q}\boldsymbol{Q}^T\boldsymbol{V}_h^T + c_h\boldsymbol{I}_p = \dot{\boldsymbol{\Sigma}}[\bar{h}],$$

as claimed. □

Remark 9. A word on the provenance of (2.5), since it is usually quoted rather than proved. It is the case $i = q$ of the general Weyl perturbation inequality for singular values,

$$|\sigma_i(\boldsymbol{A} + \boldsymbol{V}) - \sigma_i(\boldsymbol{A})| \leq \sigma_1(\boldsymbol{V}) = \|\boldsymbol{V}\|_2, \qquad i = 1, \dots, q, \tag{2.7}$$

which in [Horn and Johnson(2013)] is equation (7.3.15), stated there as Problem 7.3.P16(d) rather than as a numbered theorem. It is worth being careful about the attribution. The theorem of Weyl that carries a number in that book, namely [Horn and Johnson(2013)] Theorem 4.3.1, is the inequality

$$\lambda_i(\boldsymbol{A} + \boldsymbol{B}) \leq \lambda_{i+j}(\boldsymbol{A}) + \lambda_{n-j}(\boldsymbol{B})$$

for the *eigenvalues of Hermitian* $\boldsymbol{A}$, $\boldsymbol{B} \in M_n$, and it does not apply directly to our situation, where $\boldsymbol{A}$ and $\boldsymbol{V}_h$ are rectangular $p \times q$ matrices and neither is Hermitian. Passing from Theorem 4.3.1 to (2.7) requires the Jordan–Wielandt device of replacing a $p \times q$ matrix $\boldsymbol{B}$ by the Hermitian matrix

$$\mathcal{B} := \begin{bmatrix} \boldsymbol{0} & \boldsymbol{B} \\ \boldsymbol{B}^T & \boldsymbol{0} \end{bmatrix} \in M_{p+q},$$

whose eigenvalues are $\pm\sigma_1(\boldsymbol{B}), \dots, \pm\sigma_q(\boldsymbol{B})$ together with $p - q$ zeros, and then applying the Hermitian result to $\mathcal{A} + \mathcal{V} = \overline{\boldsymbol{A} + \boldsymbol{V}}$; this is precisely the route indicated in [Horn and Johnson(2013)], Problem 7.3.P16(a). Only the single case $i = q$ of (2.7) is needed here, and in that case, as the proof above shows, the variational characterization (2.6) gives it in three lines with no Hermitian detour; so we have preferred the direct argument and use (2.7) nowhere.

Differentiability in quadratic mean

In the classical Euclidean theory the parameter perturbation is $\theta + h$; on a manifold there is no such addition, and its role is played by $\mathrm{Exp}_{[\boldsymbol{A},s]}(h)$. Likewise the score, which classically is a vector in $\mathbb{R}^{\dim\Theta}$, becomes a linear functional on the tangent space, that is, an element of the cotangent space $T^*_{[\boldsymbol{A},s]}M$. The following definition is the manifold version of differentiability in quadratic mean; it reproduces [Sun et al.(2026)Sun, Lin, and Liu], Definition III.1, with a change of symbols, the model tangent space being their equation (7).

Definition 10. Let $[\boldsymbol{A}, s] \in M$ and abbreviate $[\boldsymbol{A}, s]_h := \mathrm{Exp}_{[\boldsymbol{A},s]}(h)$, defined for $h \in T_{[\boldsymbol{A},s]}M$ of small enough norm by Lemma 8. The model $\mathcal{P}$ of (2.2) is said to be *differentiable in quadratic mean* (DQM) at $P_{[\boldsymbol{A},s]}$ if there exists a map

$$\dot{\ell}_{[\boldsymbol{A},s]} : \mathbb{R}^p \longrightarrow T^*_{[\boldsymbol{A},s]}M,$$

called the *score operator*, assigning to each $\boldsymbol{x} \in \mathbb{R}^p$ a linear functional $h \mapsto \dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)$ on $T_{[\boldsymbol{A},s]}M$, such that, as $h \to \boldsymbol{0}$ in $T_{[\boldsymbol{A},s]}M$,

$$\int_{\mathbb{R}^p} \Big(r(\boldsymbol{x}; [\boldsymbol{A}, s]_h) - r(\boldsymbol{x}; [\boldsymbol{A}, s]) - \tfrac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)\, r(\boldsymbol{x}; [\boldsymbol{A}, s]) \Big)^2 d\boldsymbol{x} = o\left(\|h\|^2\right). \tag{2.8}$$

In that case we write $S_{[\boldsymbol{A},s]} := \dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})$ for the *score* of the model, where $\boldsymbol{X} \sim P_{[\boldsymbol{A},s]}$, and call

$$\Lambda_{[\boldsymbol{A},s]} := \overline{\operatorname{span}}\left\{S_{[\boldsymbol{A},s]}(h) \mid h \in T_{[\boldsymbol{A},s]}M\right\} \subseteq L^2\left(P_{[\boldsymbol{A},s]}\right)$$

the *model tangent space* at $[\boldsymbol{A}, s]$.

Remark 11. Three comments on the shape of (2.8). First, the norm $\|h\|$ and the space $T_{[\boldsymbol{A},s]}M$ in which h lives are intrinsic: no chart has been chosen, and the condition is invariant under the choice of representative $(\boldsymbol{A}, s)$ of the class. Second, the perturbation $[\boldsymbol{A}, s]_h$ is the geodesic one, not a coordinate increment; the whole difficulty of transporting Le Cam's theory to a manifold usually lies here, and Proposition 2 is what dissolves it in the present model, since the geodesic perturbation is affine in the horizontal lift. Third, the score operator, if it exists, is unique: (2.8) determines the element $\frac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\cdot)(h)\, r(\cdot; [\boldsymbol{A}, s])$ of $L^2(\mathbb{R}^p, d\boldsymbol{x})$ as a limit of difference quotients, and $r(\cdot; [\boldsymbol{A}, s]) > 0$ everywhere.

The score and the Fisher information operator

We now produce the candidate score operator and compute its first two moments. The bilinear form $G_{[\boldsymbol{A},s]}$ obtained below is the Fisher information operator of [Sun et al.(2026)Sun, Lin, and Liu], Definition III.2, namely the covariance $\mathbb{E}\left[S_{[\boldsymbol{A},s]}(h)\, S_{[\boldsymbol{A},s]}(h')\right]$ of the score, evaluated in our model. The following elementary Gaussian identity is the only probabilistic input.

Lemma 12. Let $\boldsymbol{\Sigma}_1 \in \mathrm{Sym}_{++}(p)$ and let $\boldsymbol{X} \sim N_p(\boldsymbol{0}, \boldsymbol{\Sigma}_1)$. For $\boldsymbol{\Delta} \in \mathrm{Sym}(p)$ define $T_{\boldsymbol{\Delta}} : \mathbb{R}^p \to \mathbb{R}$ by

$$T_{\boldsymbol{\Delta}}(\boldsymbol{x}) := \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_1^{-1}\left(\boldsymbol{x}\boldsymbol{x}^T - \boldsymbol{\Sigma}_1\right)\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\right). \tag{2.9}$$

Then $\boldsymbol{\Delta} \mapsto T_{\boldsymbol{\Delta}}$ is linear, $T_{\boldsymbol{\Delta}}(\boldsymbol{X})$ is square integrable with $\mathbb{E}\left[T_{\boldsymbol{\Delta}}(\boldsymbol{X})\right] = 0$, and for all $\boldsymbol{\Delta}, \boldsymbol{\Delta}' \in \mathrm{Sym}(p)$

$$\mathbb{E}\left[T_{\boldsymbol{\Delta}}(\boldsymbol{X})T_{\boldsymbol{\Delta}'}(\boldsymbol{X})\right] = \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}'\right). \tag{2.10}$$

Proof. Linearity is clear from (2.9). Put $\boldsymbol{M} := \boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\boldsymbol{\Sigma}_1^{-1}$, which is symmetric because $\boldsymbol{\Sigma}_1^{-1}$ and $\boldsymbol{\Delta}$ are, and likewise $\boldsymbol{M}' := \boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}'\boldsymbol{\Sigma}_1^{-1}$. Using the cyclic property of the trace twice,

$$\operatorname{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{x}\boldsymbol{x}^T\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\right) = \boldsymbol{x}^T\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\boldsymbol{\Sigma}_1^{-1}\boldsymbol{x} = \boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x}, \qquad \operatorname{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\right) = \operatorname{tr}(\boldsymbol{\Sigma}_1\boldsymbol{M}),$$

so that

$$T_{\boldsymbol{\Delta}}(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^T\boldsymbol{M}\boldsymbol{x} - \tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}_1\boldsymbol{M}). \tag{2.11}$$

Thus $T_{\boldsymbol{\Delta}}$ is a quadratic polynomial in $\boldsymbol{x}$ and is square integrable against a Gaussian law; and $\mathbb{E}\left[\boldsymbol{X}\boldsymbol{X}^T\right] = \boldsymbol{\Sigma}_1$ together with the linearity and continuity of the trace gives $\mathbb{E}\left[T_{\boldsymbol{\Delta}}(\boldsymbol{X})\right] = 0$.

For the covariance it therefore suffices, by (2.11), to prove

$$\operatorname{Cov}\left(\boldsymbol{X}^T\boldsymbol{M}\boldsymbol{X},\, \boldsymbol{X}^T\boldsymbol{M}'\boldsymbol{X}\right) = 2\operatorname{tr}\left(\boldsymbol{M}\boldsymbol{\Sigma}_1\boldsymbol{M}'\boldsymbol{\Sigma}_1\right). \tag{2.12}$$

Whiten by writing $\boldsymbol{X} = \boldsymbol{\Sigma}_1^{1/2}\boldsymbol{Z}$ with $\boldsymbol{Z} \sim N_p(\boldsymbol{0}, \boldsymbol{I}_p)$, and set $\widetilde{\boldsymbol{M}} := \boldsymbol{\Sigma}_1^{1/2}\boldsymbol{M}\boldsymbol{\Sigma}_1^{1/2}$ and $\widetilde{\boldsymbol{M}'} := \boldsymbol{\Sigma}_1^{1/2}\boldsymbol{M}'\boldsymbol{\Sigma}_1^{1/2}$, both symmetric, so that $\boldsymbol{X}^T\boldsymbol{M}\boldsymbol{X} = \boldsymbol{Z}^T\widetilde{\boldsymbol{M}}\boldsymbol{Z}$ and $\boldsymbol{X}^T\boldsymbol{M}'\boldsymbol{X} = \boldsymbol{Z}^T\widetilde{\boldsymbol{M}'}\boldsymbol{Z}$. For independent standard normal coordinates one has Isserlis' formula

$$\mathbb{E}\left[Z_iZ_jZ_kZ_l\right] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk},$$

which is checked directly: if some index occurs an odd number of times the expectation vanishes and so does the right hand side; if the four indices split into two distinct pairs both sides equal 1; and if all four coincide both sides equal 3. Hence

$$\begin{aligned}\mathbb{E}\left[\boldsymbol{Z}^T\widetilde{\boldsymbol{M}}\boldsymbol{Z}\cdot\boldsymbol{Z}^T\widetilde{\boldsymbol{M}}'\boldsymbol{Z}\right] &= \sum_{i,j,k,l}\widetilde{\boldsymbol{M}}_{ij}\widetilde{\boldsymbol{M}}'_{kl}\,\mathbb{E}\left[Z_iZ_jZ_kZ_l\right]\\ &= \left(\sum_i\widetilde{\boldsymbol{M}}_{ii}\right)\left(\sum_k\widetilde{\boldsymbol{M}}'_{kk}\right)+\sum_{i,j}\widetilde{\boldsymbol{M}}_{ij}\widetilde{\boldsymbol{M}}'_{ij}+\sum_{i,j}\widetilde{\boldsymbol{M}}_{ij}\widetilde{\boldsymbol{M}}'_{ji}\\ &= \operatorname{tr}\widetilde{\boldsymbol{M}}\ \operatorname{tr}\widetilde{\boldsymbol{M}}'+2\operatorname{tr}\left(\widetilde{\boldsymbol{M}}\widetilde{\boldsymbol{M}}'\right),\end{aligned}$$

the last equality using the symmetry of $\widetilde{\boldsymbol{M}}'$. Since $\mathbb{E}\left[\boldsymbol{Z}^T\widetilde{\boldsymbol{M}}\boldsymbol{Z}\right]=\operatorname{tr}\widetilde{\boldsymbol{M}}$ and similarly for $\widetilde{\boldsymbol{M}}'$, subtracting the product of the means leaves

$$\operatorname{Cov}\left(\boldsymbol{X}^T\boldsymbol{M}\boldsymbol{X},\,\boldsymbol{X}^T\boldsymbol{M}'\boldsymbol{X}\right)=2\operatorname{tr}\left(\widetilde{\boldsymbol{M}}\widetilde{\boldsymbol{M}}'\right)=2\operatorname{tr}\left(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{M}\boldsymbol{\Sigma}_1\boldsymbol{M}'\boldsymbol{\Sigma}_1^{1/2}\right)=2\operatorname{tr}\left(\boldsymbol{M}\boldsymbol{\Sigma}_1\boldsymbol{M}'\boldsymbol{\Sigma}_1\right),$$

which is (2.12). Substituting back,

$$\mathbb{E}\left[T_{\boldsymbol{\Delta}}(\boldsymbol{X})T_{\boldsymbol{\Delta}'}(\boldsymbol{X})\right]=\tfrac{1}{4}\cdot2\operatorname{tr}\left(\boldsymbol{M}\boldsymbol{\Sigma}_1\boldsymbol{M}'\boldsymbol{\Sigma}_1\right)=\tfrac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}'\right),$$

which is (2.10).

Neither of the two steps above is new: both are special cases of moment formulas for the matrix variate normal law collected in [Gupta and Nagar(2000)]. There a $q\times n$ random matrix satisfies $\boldsymbol{Y}\sim N_{q,n}\left(\boldsymbol{M}_*,\boldsymbol{\Sigma}_*\otimes\boldsymbol{\Psi}\right)$ when $\operatorname{vec}\left(\boldsymbol{Y}^T\right)\sim N_{qn}\left(\operatorname{vec}\left(\boldsymbol{M}_*^T\right),\boldsymbol{\Sigma}_*\otimes\boldsymbol{\Psi}\right)$ ([Gupta and Nagar(2000)], Definition 2.2.1), so the choice $q=1$, $n=p$, $\boldsymbol{M}_*=\boldsymbol{0}$, $\boldsymbol{\Sigma}_*=1$ and $\boldsymbol{\Psi}=\boldsymbol{\Sigma}_1$ makes $\boldsymbol{Y}$ the row vector $\boldsymbol{X}^T$ and turns a matrix quadratic form $\boldsymbol{Y}\boldsymbol{A}\boldsymbol{Y}^T$ into the scalar $\boldsymbol{X}^T\boldsymbol{A}\boldsymbol{X}$; we write $(\cdot)^T$ for the transpose that [Gupta and Nagar(2000)] denotes by a prime, the prime in $\boldsymbol{M}'$ here being a label and not a transpose. Under this dictionary [Gupta and Nagar(2000)], Corollary 2.3.3.1(iii), is Isserlis' formula stated directly for $\boldsymbol{\Sigma}_1$,

$$\mathbb{E}\left[X_iX_jX_kX_l\right]=(\boldsymbol{\Sigma}_1)_{ij}(\boldsymbol{\Sigma}_1)_{kl}+(\boldsymbol{\Sigma}_1)_{ik}(\boldsymbol{\Sigma}_1)_{jl}+(\boldsymbol{\Sigma}_1)_{il}(\boldsymbol{\Sigma}_1)_{jk},$$

which makes the whitening step unnecessary, while [Gupta and Nagar(2000)], Theorem 7.7.2, gives

$$\operatorname{Cov}\left(\boldsymbol{Y}\boldsymbol{A}\boldsymbol{Y}^T,\,\boldsymbol{Y}\boldsymbol{B}\boldsymbol{Y}^T\right)=\operatorname{tr}\left(\boldsymbol{A}\boldsymbol{\Psi}\boldsymbol{B}^T\boldsymbol{\Psi}\right)+\operatorname{tr}\left(\boldsymbol{A}^T\boldsymbol{\Psi}\boldsymbol{B}^T\boldsymbol{\Psi}\right)$$

once the terms carrying the mean $\boldsymbol{M}_*$ drop out and the commutation matrix reduces to $K_{11}=1$; with $\boldsymbol{A}:=\boldsymbol{M}$ and $\boldsymbol{B}:=\boldsymbol{M}'$, both symmetric, this is exactly (2.12). The same identity can be read off from [Gupta and Nagar(2000)], Theorem 7.7.1(ii), combined with the first moment $\mathbb{E}\left[\boldsymbol{Y}\boldsymbol{A}\boldsymbol{Y}^T\right]=\operatorname{tr}\left(\boldsymbol{A}^T\boldsymbol{\Psi}\right)$ of Theorem 2.3.5(ii) there. We have kept the direct derivation because it is short and self contained. □

Lemma 13. Fix $[\boldsymbol{A},s]\in M$ and write $\boldsymbol{\Sigma}_0:=\boldsymbol{\Sigma}[\boldsymbol{A},s]$. Define, for $\boldsymbol{x}\in\mathbb{R}^p$ and $h,\,h'\in T_{[\boldsymbol{A},s]}M$,

$$\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h):=\frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\left(\boldsymbol{x}\boldsymbol{x}^T-\boldsymbol{\Sigma}_0\right)\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}]\right),\tag{2.13}$$

$$G_{[\boldsymbol{A},s]}(h,h'):=\frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}]\,\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}']\right).\tag{2.14}$$

Then:

1. For each $\boldsymbol{x}$ the map $h\mapsto\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)$ is linear, so that $\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})\in T^*_{[\boldsymbol{A},s]}M$; and $G_{[\boldsymbol{A},s]}$ is a symmetric bilinear form on $T_{[\boldsymbol{A},s]}M$. Neither depends on the representative $(\boldsymbol{A},s)$ of the class.

2. If $\boldsymbol{X} \sim P_{[\boldsymbol{A},s]}$ then $\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})(h)$ is square integrable with

$$\mathbb{E}\left[\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})(h)\right] = 0, \qquad \mathbb{E}\left[\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})(h)\,\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})(h')\right] = G_{[\boldsymbol{A},s]}(h,h').$$

3. $G_{[\boldsymbol{A},s]}$ is positive semidefinite; more precisely,

$$G_{[\boldsymbol{A},s]}(h,h) = \frac{1}{2}\left\|\boldsymbol{\Sigma}_0^{-1/2}\dot{\boldsymbol{\Sigma}}[\bar{h}]\boldsymbol{\Sigma}_0^{-1/2}\right\|_F^2, \tag{2.15}$$

so that $G_{[\boldsymbol{A},s]}(h,h) = 0$ if and only if $\dot{\boldsymbol{\Sigma}}[\bar{h}] = \mathbf{0}$.

Proof. (1) By Lemma 8(1) and (2), $h \mapsto \bar{h} \mapsto \dot{\boldsymbol{\Sigma}}[\bar{h}]$ is a composition of linear maps, with values in $\mathrm{Sym}(p)$; composing with the linear map $\boldsymbol{\Delta} \mapsto T_{\boldsymbol{\Delta}}(\boldsymbol{x})$ of Lemma 12 (taken with $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_0$) gives (2.13), which is therefore linear in h. Bilinearity of (2.14) is likewise clear, and symmetry follows from the cyclic property of the trace together with the symmetry of $\boldsymbol{\Sigma}_0^{-1}$ and of $\dot{\boldsymbol{\Sigma}}[\cdot]$: writing $\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}]\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}']$ and transposing inside the trace exchanges $\bar{h}$ and $\bar{h}'$. Independence of the representative is Lemma 8(4), $\boldsymbol{\Sigma}_0$ itself being representative independent by construction.

(2) With $\boldsymbol{\Sigma}_1 := \boldsymbol{\Sigma}_0$ and $\boldsymbol{\Delta} := \dot{\boldsymbol{\Sigma}}[\bar{h}]$ we have $\dot{\ell}_{[\boldsymbol{A},s]}(\cdot)(h) = T_{\boldsymbol{\Delta}}$ in the notation (2.9), and $P_{[\boldsymbol{A},s]}$ is exactly $N_p(\mathbf{0},\boldsymbol{\Sigma}_0)$ by (2.2). The claims are then Lemma 12 verbatim, the covariance formula (2.10) being (2.14).

(3) Put $\boldsymbol{N} := \boldsymbol{\Sigma}_0^{-1/2}\dot{\boldsymbol{\Sigma}}[\bar{h}]\boldsymbol{\Sigma}_0^{-1/2}$, which is symmetric, so that $\dot{\boldsymbol{\Sigma}}[\bar{h}] = \boldsymbol{\Sigma}_0^{1/2}\boldsymbol{N}\boldsymbol{\Sigma}_0^{1/2}$ and $\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}] = \boldsymbol{\Sigma}_0^{-1/2}\boldsymbol{N}\boldsymbol{\Sigma}_0^{1/2}$. Squaring, $\left(\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\bar{h}]\right)^2 = \boldsymbol{\Sigma}_0^{-1/2}\boldsymbol{N}^2\boldsymbol{\Sigma}_0^{1/2}$, whose trace is $\mathrm{tr}(\boldsymbol{N}^2) = \|\boldsymbol{N}\|_F^2$ by cyclicity and symmetry of $\boldsymbol{N}$. This is (2.15). Finally $\|\boldsymbol{N}\|_F = 0$ forces $\boldsymbol{N} = \mathbf{0}$ and hence $\dot{\boldsymbol{\Sigma}}[\bar{h}] = \boldsymbol{\Sigma}_0^{1/2}\boldsymbol{N}\boldsymbol{\Sigma}_0^{1/2} = \mathbf{0}$, the converse being trivial. □

Remark 14. Part (3) reduces the nondegeneracy of the Fisher information operator to the purely linear algebraic question of whether $\dot{\boldsymbol{\Sigma}}[\bar{h}] = \mathbf{0}$ forces $\bar{h} = \mathbf{0}$ for *horizontal* $\bar{h}$. Equivalently, writing $V_{(\boldsymbol{A},s)}$ and $H_{(\boldsymbol{A},s)}$ as in Proposition 1, the question is whether $\ker\left(\dot{\boldsymbol{\Sigma}}\right) = V_{(\boldsymbol{A},s)}$ exactly, so that $\ker\left(\dot{\boldsymbol{\Sigma}}\right) \cap H_{(\boldsymbol{A},s)} = \{\mathbf{0}\}$. This is settled in the affirmative by Proposition 15 below, and — as that proof makes plain — with no condition whatsoever on the multiplicities of the singular values of $\boldsymbol{A}$. Almost all of the work has in fact already been done: the identification of $\ker(\dot{\boldsymbol{\Sigma}})$ was carried out inside the proof of Proposition 1, where $\dot{\boldsymbol{\Sigma}}$ appears as the differential $d\Phi_{(\boldsymbol{A},s)}$. Let us also record that nondegeneracy is *not* needed for the differentiability in quadratic mean proved below, and is not assumed there; it becomes indispensable only later, when $G_{[\boldsymbol{A},s]}^{-1}$ is to be interpreted as an asymptotic covariance.

Proposition 15. Let $[\boldsymbol{A},s] \in M$. Then $G_{[\boldsymbol{A},s]}$ is positive definite: for $h \in T_{[\boldsymbol{A},s]}M$,

$$G_{[\boldsymbol{A},s]}(h,h) = 0 \iff h = \mathbf{0}.$$

Equivalently, $\ker\left(\dot{\boldsymbol{\Sigma}}\right) = V_{(\boldsymbol{A},s)}$ exactly, and $\dot{\boldsymbol{\Sigma}}$ restricted to $H_{(\boldsymbol{A},s)}$ is injective.

Proof. The implication $\Leftarrow$ is bilinearity. For $\Rightarrow$, suppose $G_{[\boldsymbol{A},s]}(h,h) = 0$ and let $\bar{h} \in H_{(\boldsymbol{A},s)}$ be the horizontal lift of h. By Lemma 13(3) this forces

$$\dot{\boldsymbol{\Sigma}}[\bar{h}] = \mathbf{0}, \qquad \text{that is,} \qquad \bar{h} \in \ker\left(\dot{\boldsymbol{\Sigma}}\right). \tag{2.16}$$

Now $\dot{\boldsymbol{\Sigma}}$ is by Lemma 8(2) the differential at $(\boldsymbol{A},s)$ of the map (2.1), which is the map called Φ in the proof of Proposition 1; so $\dot{\boldsymbol{\Sigma}} = d\Phi_{(\boldsymbol{A},s)}$ and (2.16) reads $\bar{h} \in \ker\left(d\Phi_{(\boldsymbol{A},s)}\right)$. That proof establishes, as its central claim, that

$$\ker\left(d\Phi_{(\boldsymbol{A},s)}\right) = \ker\left(d\pi_{(\boldsymbol{A},s)}\right) = V_{(\boldsymbol{A},s)}, \tag{2.17}$$

the vertical space. Hence $\bar{h} \in V_{(\boldsymbol{A},s)}$. But $\bar{h}$ is horizontal, and $H_{(\boldsymbol{A},s)} = V_{(\boldsymbol{A},s)}^{\perp}$ by definition, so

$$\bar{h} \in V_{(\boldsymbol{A},s)} \cap V^{\perp}_{(\boldsymbol{A},s)} = \{\mathbf{0}\}\,.$$

Therefore $\bar{h} = \mathbf{0}$, and since $v \mapsto \bar{v}$ is a linear isomorphism onto $H_{(\boldsymbol{A},s)}$ we conclude $h = \mathbf{0}$. The two reformulations are immediate: (2.17) is the first, and the second follows from it together with $H_{(\boldsymbol{A},s)} \cap V_{(\boldsymbol{A},s)} = \{\mathbf{0}\}$. □

Remark 16. Two comments on the hypotheses.

First, *no eigengap is used.* Nothing in the proof refers to the singular values of $\boldsymbol{A}$, let alone to their being distinct. This is worth stressing because the classical route to asymptotic normality in principal component analysis — perturbation of the eigenvectors of the sample covariance, followed by the delta method — does require the leading population eigenvalues to be simple. The reason for the difference is that individual principal directions are not what is being estimated here: the point $[\boldsymbol{A}, s]$ of the quotient is, and the quotient has already absorbed the rotational ambiguity that repeated singular values would otherwise create.

Second, *the hypothesis $q < p$ is essential*, and is exactly where it enters: the proof of Proposition 1 produces a nonzero $\boldsymbol{u} \in \mathbb{R}^p$ orthogonal to every column of $\boldsymbol{A}$, which exists precisely because $\operatorname{rank}(\boldsymbol{A}) = q < p$, and uses it to force $c = 0$. Without it the conclusion is false. Indeed, suppose $q = p$, so that $\boldsymbol{A}$ is invertible, and take any $c \neq 0$ together with $\boldsymbol{V} := -\frac{c}{2}\boldsymbol{A}^{-T}$. Then $\boldsymbol{A}^T\boldsymbol{V} = -\frac{c}{2}\boldsymbol{I}_p$ is symmetric, so $(\boldsymbol{V}, c)$ is horizontal by Proposition 1, while

$$\dot{\boldsymbol{\Sigma}}\,[(\boldsymbol{V}, c)] = \boldsymbol{V}\boldsymbol{A}^T + \boldsymbol{A}\boldsymbol{V}^T + c\boldsymbol{I}_p = -\tfrac{c}{2}\boldsymbol{I}_p - \tfrac{c}{2}\boldsymbol{I}_p + c\boldsymbol{I}_p = \mathbf{0},$$

so this nonzero horizontal vector lies in $\ker(\dot{\boldsymbol{\Sigma}})$ and $G_{[\boldsymbol{A},s]}$ is degenerate. The smallest instance is $p = q = 1$, where $\boldsymbol{\Sigma}(a, s) = a^2 + s$ maps a two dimensional parameter space onto a one dimensional set of variances while the vertical space is trivial; the model is then hopelessly non-identifiable even after quotienting, which is the concrete reason the standing assumption $q < p$ of Section 1.1 cannot be dispensed with.

Remark 17. It is worth explaining why the conclusion of Proposition 15 deserves to be called *nondegeneracy of the model*, and how it lines up with what that word means classically. Everything below rests on one identity. Anticipating Proposition 18, and writing $d_H(P, Q) := \|r_P - r_Q\|_{L^2}$ for the Hellinger distance in the normalisation induced by (2.3), the DQM condition (2.8) says exactly that $r\,(\cdot\,; [\boldsymbol{A}, s]_h) - r\,(\cdot\,; [\boldsymbol{A}, s]) = \frac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\cdot)(h)\, r\,(\cdot\,; [\boldsymbol{A}, s]) + o(\|h\|)$ in L^2. Since $r\,(\cdot\,; [\boldsymbol{A}, s])^2$ is the density of $P_{[\boldsymbol{A},s]}$, the leading term has squared L^2 norm

$$\int_{\mathbb{R}^p} \left(\tfrac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)\right)^2 r\,(\boldsymbol{x}; [\boldsymbol{A}, s])^2\, d\boldsymbol{x} = \tfrac{1}{4}\,\mathbb{E}\left[\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})(h)^2\right] = \tfrac{1}{4}\, G_{[\boldsymbol{A},s]}(h, h)$$

by Lemma 13(2), so that

$$d_H\left(P_{[\boldsymbol{A},s]_h}, P_{[\boldsymbol{A},s]}\right)^2 = \frac{1}{4}\, G_{[\boldsymbol{A},s]}(h, h) + o\left(\|h\|^2\right). \tag{2.18}$$

So $G_{[\boldsymbol{A},s]}(h, h)$ is, up to the factor $\frac{1}{4}$, the *squared speed at which the distribution moves* when the parameter is moved in the direction h. Positive definiteness of G therefore says: no direction of the parameter space is invisible to the data. That is the content, and each of the classical meanings of nondegeneracy is a rephrasing of it.

Local identifiability. The model (2.2) is identifiable by construction, $[\boldsymbol{A}, s] \mapsto P_{[\boldsymbol{A},s]}$ being injective. It is essential to see that this does *not* already give Proposition 15: injectivity of a smooth map does not imply injectivity of its differential, as $t \mapsto t^3$ on $\mathbb{R}$ shows. Identifiability is a statement about the map, nondegeneracy about its derivative, and the latter is strictly stronger locally. Concretely, if $G(h, h) = 0$ for some $h \neq \mathbf{0}$ then by (2.18) the parameter can be moved at unit speed along the geodesic $t \mapsto [\boldsymbol{A}, s]_{th}$ while the distribution stays put to first order: distinct parameters, but statistically indistinguishable at the relevant scale.

The classical regularity condition. In the Euclidean theory one says the model is regular at ϑ_0 when it is DQM there *and* the Fisher information matrix I_{ϑ_0} is nonsingular. Those two conditions are, in the present setting, exactly Proposition 18 and Proposition 15, so with the latter in hand we have established the full classical regularity hypothesis on M. Singularity of I_{ϑ_0} is what classically renders the Cramér–Rao bound vacuous and the limiting Gaussian shift experiment $\left\{N\left(h, I_{\vartheta_0}^{-1}\right)\right\}$ meaningless; here it is what would prevent the covariance $G_{[\boldsymbol{A},s]}^{-1}$, which is to be the asymptotic covariance of the estimator, from existing at all.

The sharpest form. Suppose $G_{[\boldsymbol{A},s]}(h,h) = 0$ for some $h \neq \mathbf{0}$. Since $G_{[\boldsymbol{A},s]}(h,h) = \mathbb{E}\left[\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})(h)^2\right]$ by Lemma 13(2), the score in the direction h vanishes almost surely. The local asymptotic normality expansion in the direction h, whose two terms are the score sum and $-\frac{1}{2}G(h,h)$, would then reduce to $o_P(1)$: the log-likelihood ratio between the perturbed and unperturbed experiments converges to zero, so the two are asymptotically indistinguishable and no test can have asymptotic power against the alternative h. Degeneracy is thus not a technical inconvenience but the assertion that a genuine direction of the parameter space carries no information whatsoever.

The quotient as a cure. There is a purely linear algebraic way to see what passing to M accomplished. A positive semidefinite form on a vector space E descends to a positive *definite* form on the quotient $E/\ker$, and this is the standard repair for a degenerate quadratic form. Formula (2.14) makes sense for an arbitrary, not necessarily horizontal, tangent vector of Θ, and so defines a Fisher form on $T_{(\boldsymbol{A},s)}\Theta$; by (2.15) and (2.17) that form is positive semidefinite with kernel exactly $V_{(\boldsymbol{A},s)}$, of dimension $\frac{q(q-1)}{2}$. Hence the model on Θ *is* degenerate in the classical sense, and every classical theorem is simply inapplicable there. Passing to the quotient is precisely the operation of dividing out the null space of the Fisher form, since $T_{(\boldsymbol{A},s)}\Theta/V_{(\boldsymbol{A},s)} \cong H_{(\boldsymbol{A},s)} \cong T_{[\boldsymbol{A},s]}M$; and the content of Proposition 15 is that the division is exactly right, neither too coarse nor too fine. This is the precise sense in which the quotient manifold M, and not Θ, is the correct parameter space for an asymptotic theory.

A metric restatement. Combining (2.18) with Proposition 15 and the compactness of the unit sphere of $T_{[\boldsymbol{A},s]}M$, there are constants $0 < c \leq C < \infty$ and $\delta > 0$ with

$$c\,\|h\| \;\leq\; d_H\left(P_{[\boldsymbol{A},s]_h},\, P_{[\boldsymbol{A},s]}\right) \;\leq\; C\,\|h\|, \qquad \|h\| < \delta.$$

That is, the statistical distance between two nearby members of the model is comparable to the Riemannian distance between the corresponding points of M. The two inequalities fail in opposite ways and both are fatal. The upper bound is the DQM half: it fails for the uniform family of the examples at the end of Section 2.1, where $d_H \asymp \|h\|^{1/2} \gg \|h\|$ and the model moves too fast, which is why estimation there is n-consistent rather than $\sqrt{n}$-consistent. The lower bound is the nondegeneracy half: it fails when G is singular, the model then moving too slowly in some direction for any $\sqrt{n}$-consistent estimator to exist there. The regular theory lives exactly between the two.

The model is differentiable in quadratic mean

Proposition 18. The model $\mathcal{P}$ of (2.2) is differentiable in quadratic mean, in the sense of Definition 10, at $P_{[\boldsymbol{A},s]}$ for *every* $[\boldsymbol{A},s] \in M$, with score operator $\dot{\ell}_{[\boldsymbol{A},s]}$ given by (2.13). Consequently the score $S_{[\boldsymbol{A},s]} = \dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{X})$ is centred and square integrable, its covariance — the Fisher information operator of the model at $[\boldsymbol{A},s]$ — is the bilinear form $G_{[\boldsymbol{A},s]}$ of (2.14), and the model tangent space is the finite dimensional subspace

$$\Lambda_{[\boldsymbol{A},s]} = \left\{\dot{\ell}_{[\boldsymbol{A},s]}(\cdot)(h) \mid h \in T_{[\boldsymbol{A},s]}M\right\} \subseteq L^2\left(P_{[\boldsymbol{A},s]}\right), \qquad \dim \Lambda_{[\boldsymbol{A},s]} \leq \dim M.$$

Proof. Fix a representative $(\boldsymbol{A},s) \in \Theta$ of the base point, which is legitimate since by Lemma 8(4) and Lemma 13(1) nothing in the statement depends on it. Abbreviate $\boldsymbol{\Sigma}_0 := \boldsymbol{\Sigma}(\boldsymbol{A},s)$, $\rho := \min\left\{\sigma_{\min}(\boldsymbol{A}), s\right\} > 0$, and $d := \dim M$. Throughout, h ranges over the open ball $B := \left\{h \in T_{[\boldsymbol{A},s]}M \mid \|h\| < \rho/2\right\}$, on which Lemma 8 applies. Write, for $h \in B$ and $\boldsymbol{x} \in \mathbb{R}^p$,

$$L(h;\boldsymbol{x}) := \log f\left(\boldsymbol{x}; [\boldsymbol{A},s]_h\right) = -\frac{p}{2}\log(2\pi) - \frac{1}{2}\log\det \boldsymbol{\Sigma}(h) - \frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}(h)^{-1}\boldsymbol{x}, \tag{2.19}$$

and $r(h;\boldsymbol{x}) := r\left(\boldsymbol{x}; [\boldsymbol{A},s]_h\right) = \exp\left(\frac{1}{2}L(h;\boldsymbol{x})\right)$, where $\boldsymbol{\Sigma}(h) \in \mathrm{Sym}_{++}(p)$ is as in Lemma 8(2)–(3). Note that $r(\mathbf{0};\cdot) = r(\cdot;[\boldsymbol{A},s])$.

We stress at the outset the structural point on which the whole proof rests: by Proposition 2 the geodesic perturbation $[\boldsymbol{A}, s]_h$ is *affine* in the horizontal lift $\bar{h}$, so that $\left\{P_{[\boldsymbol{A},s]_h} \mid h \in B\right\}$ is an ordinary parametric family indexed by an open subset of the d dimensional Euclidean space $H_{(\boldsymbol{A},s)} \cong T_{[\boldsymbol{A},s]}M$. No approximation of the exponential map is involved anywhere below; the manifold statement (2.8) *is* the classical quadratic mean differentiability of this Euclidean family at $h = \mathbf{0}$, that is, condition (7.1) of [van der Vaart(1998)], page 93. One could therefore finish by quoting a standard sufficient condition, namely [van der Vaart(1998)], Lemma 7.6, which asks that $h \mapsto \sqrt{f\left(\boldsymbol{x}; [\boldsymbol{A}, s]_h\right)}$ be continuously differentiable for every $\boldsymbol{x}$ and that the Fisher information be well defined and continuous in h; Steps 1 to 4 below verify precisely these two hypotheses. We nevertheless carry the argument through to the end in Step 5, both to keep the section self-contained and because the passage from the directionwise statement to the uniform one, invisible in the one dimensional phrasing of the quoted lemma, is the only genuinely delicate point.

Step 1: pointwise smoothness in h and the derivative. By (2.4) the map $h \mapsto \boldsymbol{\Sigma}(h)$ is a polynomial of degree two with values in $\mathrm{Sym}(p)$, hence C^∞, and since $h \mapsto \bar{h}$ is linear, its directional derivative at $h \in B$ in the direction $v \in T_{[\boldsymbol{A},s]}M$ is, writing $\bar{v} = (\boldsymbol{U}, b)$,

$$D_v\boldsymbol{\Sigma}(h) = \boldsymbol{U}\left(\boldsymbol{A} + \boldsymbol{V}_h\right)^T + \left(\boldsymbol{A} + \boldsymbol{V}_h\right)\boldsymbol{U}^T + b\boldsymbol{I}_p =: \dot{\boldsymbol{\Sigma}}_h[\bar{v}] \in \mathrm{Sym}(p), \tag{2.20}$$

obtained by differentiating $t \mapsto \boldsymbol{\Sigma}(h + tv)$ at $t = 0$ and using $\overline{h + tv} = \bar{h} + t\bar{v}$. In particular $\dot{\boldsymbol{\Sigma}}_{\mathbf{0}}[\bar{v}] = \dot{\boldsymbol{\Sigma}}[\bar{v}]$, and $v \mapsto \dot{\boldsymbol{\Sigma}}_h[\bar{v}]$ is linear for each fixed h.

On the open cone $\mathrm{Sym}_{++}(p)$ the maps $\boldsymbol{\Sigma} \mapsto \log\det\boldsymbol{\Sigma}$ and $\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ are C^∞, with differentials $\boldsymbol{\Delta} \mapsto \mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\boldsymbol{\Delta}\right)$ and $\boldsymbol{\Delta} \mapsto -\boldsymbol{\Sigma}^{-1}\boldsymbol{\Delta}\boldsymbol{\Sigma}^{-1}$ respectively. By Lemma 8(3), $\boldsymbol{\Sigma}(h)$ stays in that cone for $h \in B$. Therefore, for every fixed $\boldsymbol{x}$, the map $h \mapsto L(h; \boldsymbol{x})$ is C^∞ on B, hence so is $h \mapsto r(h; \boldsymbol{x}) = \exp\left(\frac{1}{2}L(h; \boldsymbol{x})\right)$, and the chain rule gives

$$\begin{aligned} D_v L(h; \boldsymbol{x}) &= -\frac{1}{2}\,\mathrm{tr}\left(\boldsymbol{\Sigma}(h)^{-1}\dot{\boldsymbol{\Sigma}}_h[\bar{v}]\right) + \frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}(h)^{-1}\dot{\boldsymbol{\Sigma}}_h[\bar{v}]\boldsymbol{\Sigma}(h)^{-1}\boldsymbol{x} \\ &= \frac{1}{2}\,\mathrm{tr}\left(\boldsymbol{\Sigma}(h)^{-1}\left(\boldsymbol{x}\boldsymbol{x}^T - \boldsymbol{\Sigma}(h)\right)\boldsymbol{\Sigma}(h)^{-1}\dot{\boldsymbol{\Sigma}}_h[\bar{v}]\right), \end{aligned} \tag{2.21}$$

where the second equality uses the cyclic property of the trace exactly as in (2.11), and

$$D_v r(h; \boldsymbol{x}) = \tfrac{1}{2}\, D_v L(h; \boldsymbol{x})\, r(h; \boldsymbol{x}). \tag{2.22}$$

Comparing (2.21) at $h = \mathbf{0}$ with (2.13) we see that

$$D_v L(\mathbf{0}; \boldsymbol{x}) = \dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(v), \tag{2.23}$$

so the candidate score of Lemma 13 is indeed the pointwise derivative of the log-likelihood in the chart. It is convenient to abbreviate, for $h \in B$ and $v \in T_{[\boldsymbol{A},s]}M$,

$$\phi_h^v(\boldsymbol{x}) := D_v L(h; \boldsymbol{x})\, r(h; \boldsymbol{x}), \qquad \text{so that} \qquad D_v r(h; \cdot) = \tfrac{1}{2}\phi_h^v \quad \text{and} \quad \phi_{\mathbf{0}}^v = \dot{\ell}_{[\boldsymbol{A},s]}(\cdot)(v)\, r(\cdot; [\boldsymbol{A}, s]). \tag{2.24}$$

By linearity of $v \mapsto \dot{\boldsymbol{\Sigma}}_h[\bar{v}]$ and of the trace, $v \mapsto \phi_h^v$ is linear for each fixed h.

Step 2: ϕ_h^v lies in $L^2(\mathbb{R}^p, d\boldsymbol{x})$, with a closed form norm. Fix $h \in B$ and v. Since $r(h; \cdot)^2$ is the $N_p(\mathbf{0}, \boldsymbol{\Sigma}(h))$ density,

$$\int_{\mathbb{R}^p} \phi_h^v(\boldsymbol{x})^2\, d\boldsymbol{x} = \int_{\mathbb{R}^p} \left(D_v L(h; \boldsymbol{x})\right)^2 f\left(\boldsymbol{x}; [\boldsymbol{A}, s]_h\right)\, d\boldsymbol{x} = \mathbb{E}\left[T_{\boldsymbol{\Delta}}(\boldsymbol{X}_h)^2\right],$$

where $\boldsymbol{X}_h \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}(h))$, and where we applied (2.21) to recognise $D_v L(h; \cdot) = T_{\boldsymbol{\Delta}}$ in the notation (2.9) of Lemma 12 taken with $\boldsymbol{\Sigma}_1 := \boldsymbol{\Sigma}(h)$ and $\boldsymbol{\Delta} := \dot{\boldsymbol{\Sigma}}_h[\bar{v}]$. Note that Lemma 12 is applied here with a *general* symmetric $\boldsymbol{\Delta}$; no horizontality of $\bar{v}$ relative to the shifted representative $(\boldsymbol{A}+\boldsymbol{V}_h, s+c_h)$ is needed, and indeed none is available. That lemma therefore gives $\phi_h^v \in L^2(\mathbb{R}^p, d\boldsymbol{x})$ with

$$\|\phi_h^v\|_{L^2}^2 = \frac{1}{2}\operatorname{tr}\left(\boldsymbol{\Sigma}(h)^{-1}\dot{\boldsymbol{\Sigma}}_h[\bar{v}]\,\boldsymbol{\Sigma}(h)^{-1}\dot{\boldsymbol{\Sigma}}_h[\bar{v}]\right) =: I(h)(v,v), \tag{2.25}$$

the Fisher information of the chart family at h; by (2.14) and (2.23), $I(\mathbf{0})(v,v) = G_{[\boldsymbol{A},s]}(v,v) = \|\phi_{\mathbf{0}}^v\|_{L^2}^2$.

Step 3: $h \mapsto \phi_h^v$ is L^2 continuous at $h = \mathbf{0}$, for each fixed v. Let $(h_n)_{n\geq 1}$ be any sequence in B with $h_n \to \mathbf{0}$.

(a) Pointwise convergence. The map $h \mapsto \boldsymbol{\Sigma}(h)$ is continuous by (2.4), takes values in $\mathrm{Sym}_{++}(p)$ by Lemma 8(3), and both $\boldsymbol{\Sigma} \mapsto \boldsymbol{\Sigma}^{-1}$ and $\boldsymbol{\Sigma} \mapsto \det\boldsymbol{\Sigma}$ are continuous there; also $h \mapsto \dot{\boldsymbol{\Sigma}}_h[\bar{v}]$ is affine by (2.20). Reading off (2.19), (2.21) and (2.24), the function $\phi_h^v(\boldsymbol{x})$ is built from these by sums, products, traces and the exponential, so that for *every* $\boldsymbol{x} \in \mathbb{R}^p$,

$$\phi_{h_n}^v(\boldsymbol{x}) \longrightarrow \phi_{\mathbf{0}}^v(\boldsymbol{x}), \qquad n \to \infty.$$

(b) Convergence of the norms. By the same continuity applied to the closed form (2.25),

$$\left\|\phi_{h_n}^v\right\|_{L^2}^2 = I(h_n)(v,v) \longrightarrow I(\mathbf{0})(v,v) = \|\phi_{\mathbf{0}}^v\|_{L^2}^2.$$

(c) From (a) and (b) to L^2 convergence. This is Riesz's criterion, which we include for completeness. Since $(\alpha-\beta)^2 \leq 2\alpha^2 + 2\beta^2$ for reals, the functions

$$\psi_n := 2\left(\phi_{h_n}^v\right)^2 + 2\left(\phi_{\mathbf{0}}^v\right)^2 - \left(\phi_{h_n}^v - \phi_{\mathbf{0}}^v\right)^2$$

are nonnegative, and by (a) they converge pointwise to $4\left(\phi_{\mathbf{0}}^v\right)^2$. Fatou's lemma and then (b) give

$$\begin{aligned}
4\|\phi_{\mathbf{0}}^v\|_{L^2}^2 &= \int_{\mathbb{R}^p} \liminf_n \psi_n\, d\boldsymbol{x} \;\leq\; \liminf_n \int_{\mathbb{R}^p} \psi_n\, d\boldsymbol{x} \\
&= \liminf_n \left(2\left\|\phi_{h_n}^v\right\|_{L^2}^2 + 2\|\phi_{\mathbf{0}}^v\|_{L^2}^2 - \left\|\phi_{h_n}^v - \phi_{\mathbf{0}}^v\right\|_{L^2}^2\right) \\
&= 4\|\phi_{\mathbf{0}}^v\|_{L^2}^2 - \limsup_n \left\|\phi_{h_n}^v - \phi_{\mathbf{0}}^v\right\|_{L^2}^2,
\end{aligned}$$

the last equality because the first two terms form a convergent sequence. As $\|\phi_{\mathbf{0}}^v\|_{L^2} < \infty$ by Step 2, we may cancel it and conclude $\limsup_n \left\|\phi_{h_n}^v - \phi_{\mathbf{0}}^v\right\|_{L^2}^2 \leq 0$, that is, $\phi_{h_n}^v \to \phi_{\mathbf{0}}^v$ in $L^2(\mathbb{R}^p, d\boldsymbol{x})$.

The sequence being arbitrary, the nondecreasing function

$$\omega_v(\delta) := \sup_{\|h\|\leq\delta} \|\phi_h^v - \phi_{\mathbf{0}}^v\|_{L^2}, \qquad 0 < \delta < \rho/2, \tag{2.26}$$

satisfies $\omega_v(\delta) \to 0$ as $\delta \downarrow 0$. Indeed, being nondecreasing it has a limit $\lambda \geq 0$ at 0^+; if $\lambda > 0$ we could choose, for each n large, some h_n with $\|h_n\| \leq 1/n$ and $\left\|\phi_{h_n}^v - \phi_{\mathbf{0}}^v\right\|_{L^2} > \lambda/2$, contradicting what we just proved.

Step 4: uniformity over directions. Let $e_1, \ldots, e_d$ be a $g_{[\boldsymbol{A},s]}$ orthonormal basis of $T_{[\boldsymbol{A},s]}M$. By Step 1 the map $v \mapsto \phi_h^v$ is linear for each fixed h, and so therefore is $v \mapsto \phi_h^v - \phi_{\mathbf{0}}^v$. Hence for a unit vector $u = \sum_{j=1}^d u_j e_j$, for which $\sum_j u_j^2 = 1$ and so $\sum_j |u_j| \leq \sqrt{d}$ by the Cauchy–Schwarz inequality, the triangle inequality in L^2 gives

$$\|\phi_h^u - \phi_{\mathbf{0}}^u\|_{L^2} = \left\|\sum_{j=1}^d u_j\left(\phi_h^{e_j} - \phi_{\mathbf{0}}^{e_j}\right)\right\|_{L^2} \leq \sum_{j=1}^d |u_j|\left\|\phi_h^{e_j} - \phi_{\mathbf{0}}^{e_j}\right\|_{L^2} \leq \sqrt{d}\max_{1\leq j\leq d}\left\|\phi_h^{e_j} - \phi_{\mathbf{0}}^{e_j}\right\|_{L^2}.$$

Taking suprema and invoking Step 3 at each of the finitely many basis vectors, the modulus of continuity

$$\omega(\delta) := \sup_{\|h\|\leq\delta}\sup_{\|u\|=1} \|\phi_h^u - \phi_{\mathbf{0}}^u\|_{L^2} \;\leq\; \sqrt{d}\max_{1\leq j\leq d}\omega_{e_j}(\delta) \tag{2.27}$$

is nondecreasing and satisfies $\omega(\delta) \to 0$ as $\delta \downarrow 0$. This is where the finite dimensionality of M enters the analysis: it is what upgrades the directionwise continuity of Step 3 to the uniformity over directions that (2.8) demands, and the argument would have to be run differently on an infinite dimensional parameter manifold.

Step 5: the fundamental theorem of calculus and Minkowski's inequality. Let $h \in B$ with $h \neq \mathbf{0}$ and set $u := h/\|h\|$, a unit vector. For t in the open interval $(-\rho/(2\|h\|), \rho/(2\|h\|)) \supset [0,1]$ the point th lies in B, so that $t \mapsto r(th;\boldsymbol{x})$ is defined there, and it is C^1 — indeed C^∞ — for each fixed $\boldsymbol{x}$, because $h' \mapsto r(h';\boldsymbol{x})$ is so by Step 1. We claim that

$$\frac{d}{dt} r(th;\boldsymbol{x}) \overset{(a)}{=} D_h r(th;\boldsymbol{x}) \overset{(b)}{=} \|h\|\, D_u r(th;\boldsymbol{x}) \overset{(c)}{=} \frac{\|h\|}{2}\, \phi^u_{th}(\boldsymbol{x}). \tag{2.28}$$

For (a), recall that D_v denotes the directional derivative in the direction v *at the point indicated by its argument*, that is

$$D_v r(h';\boldsymbol{x}) := \frac{d}{ds}\Big|_{s=0} r(h'+sv;\boldsymbol{x}). \tag{2.29}$$

This is meaningful because the parameter h' ranges over an open subset of the *vector space* $T_{[\boldsymbol{A},s]}M$, in which a point and a direction are the same kind of object and the sum $h' + sv$ makes sense; we are working inside the exponential chart precisely so that this is so. Now observe that the curve $t \mapsto th$ is a straight line traversed at the constant velocity h, so that incrementing its parameter t by s increments its value by exactly sh, and hence

$$\frac{d}{dt} r(th;\boldsymbol{x}) = \lim_{s\to 0} \frac{r((t+s)h;\boldsymbol{x}) - r(th;\boldsymbol{x})}{s} = \lim_{s\to 0} \frac{r(th+sh;\boldsymbol{x}) - r(th;\boldsymbol{x})}{s} = D_h r(th;\boldsymbol{x}),$$

the last expression being, by (2.29) with $h' = th$ and $v = h$, exactly the defining difference quotient of D_h evaluated at the point th. The two derivatives in (a) are thus the same limit written twice. This is of course the chain rule applied to $r(\cdot\,;\boldsymbol{x}) \circ \gamma$ with $\gamma(t) = th$ and $\gamma'(t) \equiv h$, but because γ is affine no chain rule is actually needed: the middle equality above is the identity $(t+s)h = th + sh$ and nothing more. Note also that the point at which the derivative is taken moves with t, whereas the direction h does not; this is what makes ϕ^u_{th}, rather than $\phi^u_{\mathbf{0}}$, appear on the right of (2.28).

Equality (b) is linearity in the direction. By (2.21) the direction v enters $D_v L(h';\boldsymbol{x})$ only through the linear map $v \mapsto \dot{\boldsymbol{\Sigma}}_{h'}[\bar{v}]$, so $v \mapsto D_v L(h';\boldsymbol{x})$ is linear, as was noted at the end of Step 1; by (2.22) the factor $r(h';\boldsymbol{x})$ does not depend on v, so $v \mapsto D_v r(h';\boldsymbol{x}) = \frac{1}{2} D_v L(h';\boldsymbol{x})\, r(h';\boldsymbol{x})$ is linear as well. Applying this at $h' = th$ to the decomposition $h = \|h\|\, u$ gives (b). Finally (c) is (2.22) combined with the definition (2.24) of ϕ^u_{th}, which say together that $D_u r(h';\cdot) = \frac{1}{2}\phi^u_{h'}$.

Integrating (2.28) from 0 to 1,

$$r(h;\boldsymbol{x}) - r(\mathbf{0};\boldsymbol{x}) = \frac{\|h\|}{2} \int_0^1 \phi^u_{th}(\boldsymbol{x})\, dt.$$

On the other hand $\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h) = \|h\|\, \dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(u)$ by linearity, so that by (2.24)

$$\tfrac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)\, r(\boldsymbol{x};[\boldsymbol{A},s]) = \frac{\|h\|}{2}\, \phi^u_{\mathbf{0}}(\boldsymbol{x}) = \frac{\|h\|}{2} \int_0^1 \phi^u_{\mathbf{0}}(\boldsymbol{x})\, dt.$$

Subtracting, the DQM remainder is exactly

$$R_h(\boldsymbol{x}) := r(\boldsymbol{x};[\boldsymbol{A},s]_h) - r(\boldsymbol{x};[\boldsymbol{A},s]) - \tfrac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)\, r(\boldsymbol{x};[\boldsymbol{A},s]) = \frac{\|h\|}{2} \int_0^1 \left(\phi^u_{th}(\boldsymbol{x}) - \phi^u_{\mathbf{0}}(\boldsymbol{x})\right) dt. \tag{2.30}$$

Abbreviate the integrand of (2.30) by

$$F(t,\boldsymbol{x}) := \phi^u_{th}(\boldsymbol{x}) - \phi^u_{\mathbf{0}}(\boldsymbol{x}), \qquad (t,\boldsymbol{x}) \in [0,1] \times \mathbb{R}^p,$$

so that (2.30) reads $R_h = \frac{\|h\|}{2}\int_0^1 F(t,\cdot)\,dt$; here h, and with it the unit vector $u = h/\|h\|$, is fixed, and only t and $\boldsymbol{x}$ vary. Recall Minkowski's integral inequality in the form we need: if (T,ν) and (Y,μ) are σ-finite measure spaces and F is $\nu\otimes\mu$ measurable, then

$$\left(\int_Y \left|\int_T F(t,\boldsymbol{y})\,d\nu(t)\right|^2 d\mu(\boldsymbol{y})\right)^{1/2} \le \int_T \left(\int_Y |F(t,\boldsymbol{y})|^2 d\mu(\boldsymbol{y})\right)^{1/2} d\nu(t), \tag{2.31}$$

that is, $\left\|\int_T F(t,\cdot)\,d\nu(t)\right\|_{L^2(\mu)} \le \int_T \|F(t,\cdot)\|_{L^2(\mu)}\,d\nu(t)$. This is nothing but the triangle inequality with the finite sum replaced by an integral: the norm of an average is at most the average of the norms. Equivalently, it is the bound $\left\|\int_T G\right\| \le \int_T \|G\|$ for the Bochner integral of the $L^2(\mu)$ valued map $G(t) := F(t,\cdot)$; we state it in the scalar form (2.31) to avoid having to set up vector valued integration.

We apply (2.31) with $T = [0,1]$ and $Y = \mathbb{R}^p$, both carrying Lebesgue measure and both σ-finite. Its hypotheses are met:

1. *Joint measurability.* By Step 1 the map $(h',\boldsymbol{x}) \mapsto \phi^u_{h'}(\boldsymbol{x}) = D_u L(h';\boldsymbol{x})\,r(h';\boldsymbol{x})$ is jointly continuous on $B\times\mathbb{R}^p$, being assembled from the continuous maps $h' \mapsto \boldsymbol{\Sigma}(h')^{-1}$ and $h' \mapsto \dot{\boldsymbol{\Sigma}}_{h'}[\bar{u}]$, the polynomial $\boldsymbol{x} \mapsto \boldsymbol{x}\boldsymbol{x}^T$, and the exponential. Precomposing with the continuous $t \mapsto th$ and subtracting the (t-independent, continuous) function $\phi^u_{\mathbf{0}}$ shows that F is jointly continuous, hence jointly measurable.

2. *The right hand side of* (2.31) *is well defined.* The function $|F|^2$ is nonnegative and jointly measurable by (1), so Tonelli's theorem makes $t \mapsto \int_{\mathbb{R}^p} |F(t,\boldsymbol{x})|^2 d\boldsymbol{x} = \|F(t,\cdot)\|^2_{L^2}$ a measurable function of t.

3. *Finiteness.* Each $F(t,\cdot)$ lies in $L^2(\mathbb{R}^p, d\boldsymbol{x})$, since ϕ^u_{th} and $\phi^u_{\mathbf{0}}$ do by Step 2.

Inequality (2.31) therefore gives

$$\|R_h\|_{L^2} = \frac{\|h\|}{2}\left\|\int_0^1 F(t,\cdot)\,dt\right\|_{L^2} \le \frac{\|h\|}{2}\int_0^1 \|\phi^u_{th} - \phi^u_{\mathbf{0}}\|_{L^2} dt. \tag{2.32}$$

It remains to bound the integrand of (2.32) uniformly in t, and this is exactly what Step 4 was arranged to provide. For every $t \in [0,1]$ we have $\|th\| = t\|h\| \le \|h\|$, while u is a unit vector; so the pair (th, u) is one of those over which the supremum defining $\omega(\|h\|)$ in (2.27) is taken, whence

$$\|\phi^u_{th} - \phi^u_{\mathbf{0}}\|_{L^2} \le \omega(\|h\|) \qquad \text{for all } t \in [0,1].$$

The integrand of (2.32) is thus dominated by a constant, and $\int_0^1 \omega(\|h\|)\,dt = \omega(\|h\|)$, so that

$$\|R_h\|_{L^2} \le \frac{\|h\|}{2}\,\omega(\|h\|).$$

Squaring,

$$\int_{\mathbb{R}^p}\left(r(\boldsymbol{x};[\boldsymbol{A},s]_h) - r(\boldsymbol{x};[\boldsymbol{A},s]) - \tfrac{1}{2}\dot{\ell}_{[\boldsymbol{A},s]}(\boldsymbol{x})(h)\,r(\boldsymbol{x};[\boldsymbol{A},s])\right)^2 d\boldsymbol{x} \le \frac{\|h\|^2}{4}\,\omega(\|h\|)^2,$$

and since $\omega(\|h\|) \to 0$ as $h \to \mathbf{0}$ by Step 4, the right hand side is $o\left(\|h\|^2\right)$. This is precisely (2.8), so $\mathcal{P}$ is differentiable in quadratic mean at $P_{[\boldsymbol{A},s]}$ with score operator $\dot{\ell}_{[\boldsymbol{A},s]}$.

Step 6: the remaining assertions. The base point $[\boldsymbol{A},s] \in M$ was arbitrary, so DQM holds everywhere on M; observe that the only properties of $(\boldsymbol{A},s)$ used were $\operatorname{rank}(\boldsymbol{A}) = q$ and $s > 0$, which hold throughout Θ by definition. That $S_{[\boldsymbol{A},s]}$ is centred and square integrable with covariance $G_{[\boldsymbol{A},s]}$ is Lemma 13(2). Finally, by Lemma 13(1) the map $h \mapsto \dot{\ell}_{[\boldsymbol{A},s]}(\cdot)(h)$ is linear from $T_{[\boldsymbol{A},s]}M$ into $L^2\left(P_{[\boldsymbol{A},s]}\right)$, so its range is a linear subspace of dimension at most $\dim T_{[\boldsymbol{A},s]}M = \dim M$. A finite dimensional subspace of a normed space is closed, so the range already equals its own closure and no completion is needed in the definition of $\Lambda_{[\boldsymbol{A},s]}$. □

Remark 19. It is worth isolating what made the proof go through, because it is special to this model and not a general phenomenon on quotient manifolds. Quadratic mean differentiability is a statement about a *one parameter family of perturbations* of the true measure, and on a manifold the natural perturbation is geodesic. In general one must then control the difference between the geodesic perturbation and any coordinate perturbation, which costs an extra approximation argument. Here Proposition 2 says that the geodesics of M are the projections of straight lines in Θ, so that $h \mapsto [\boldsymbol{A}, s]_h$ is affine in the horizontal lift; and Lemma 8(2) says that the covariance, read in that chart, is an exact quadratic polynomial in h with no remainder. The family $\left\{P_{[\boldsymbol{A},s]_h}\right\}$ is then a perfectly ordinary Euclidean exponential type family, and the only genuine analysis left is the L^2 continuity of Step 3. By contrast, positive definiteness of $G_{[\boldsymbol{A},s]}$ (Proposition 15) is where the quotient structure really bites: it is the one statement in this section that would be false on Θ.

2.1 The Geometric Content of Differentiability in Quadratic Mean

Definition 10 is short but opaque on first reading: it is not obvious where the square root comes from, why the perturbation is squared and integrated, why the factor $\frac{1}{2}$ appears, or what the condition is really asking of the model. This section answers those questions. The short answer is that *differentiability in quadratic mean is nothing other than Fréchet differentiability, at the point in question, of the map sending a parameter to the square root of its density, regarded as a map into the Hilbert space L^2*. Everything else in the definition is bookkeeping around that one idea.

Most of this section is expository and is not used later. Two points are not: the warning of Remark 21 that M carries two distinct and non-proportional bilinear forms, and the factorization of Proposition 22, which is the cleanest way to see what Proposition 15 really says.

The root density map

Let us first strip away the manifold and recall the classical picture. Let $U \subseteq \mathbb{R}^k$ be open, let μ be a measure on a sample space $\mathcal{X}$, and let $\{p_\vartheta \mid \vartheta \in U\}$ be a family of μ-densities. Put $\mathcal{H} := L^2(\mathcal{X}, \mu)$, a real Hilbert space, and define

$$\Phi : U \longrightarrow \mathcal{H}, \qquad \Phi(\vartheta) := r_\vartheta = \sqrt{p_\vartheta}. \tag{2.33}$$

The classical DQM condition at ϑ_0 reads

$$\int_{\mathcal{X}} \left(r_{\vartheta_0+h} - r_{\vartheta_0} - \tfrac{1}{2}\dot{\ell}_{\vartheta_0}(h)\, r_{\vartheta_0} \right)^2 d\mu = o\left(\|h\|^2\right), \tag{2.34}$$

which, since the integral of a square is exactly a squared $\mathcal{H}$-norm, is the same as

$$\|\Phi(\vartheta_0 + h) - \Phi(\vartheta_0) - D\Phi_{\vartheta_0}(h)\|_{\mathcal{H}} = o\left(\|h\|\right), \qquad D\Phi_{\vartheta_0}(h) := \tfrac{1}{2}\dot{\ell}_{\vartheta_0}(h)\, r_{\vartheta_0}. \tag{2.35}$$

That is precisely the definition of Fréchet differentiability of Φ at ϑ_0 with derivative the bounded linear map $D\Phi_{\vartheta_0}$. The same remark applies verbatim to (2.8), with U replaced by M and μ by Lebesgue measure on $\mathbb{R}^p$.

Remark 20. It is worth saying explicitly that (2.34) is a *first order* condition, notwithstanding the exponent 2 on both sides. The square on the left is the square of the Hilbert norm and the square on the right compensates for it; after taking square roots one is left with (2.35), an ordinary first order Taylor statement. The condition is stated in the squared form only because the left hand side is then an integral one can actually estimate.

Three features of the definition now explain themselves.

Why the square root. Because $\int r_\vartheta^2 \, d\mu = \int p_\vartheta \, d\mu = 1$, the map Φ takes its values in the *unit sphere*

$$S(\mathcal{H}) := \{u \in \mathcal{H} \mid \|u\|_{\mathcal{H}} = 1\},$$

a fixed subset of a fixed Hilbert space, the same one for every ϑ. The normalisation constraint on a density, which is an awkward affine side condition if one works with p_ϑ itself, becomes a piece of geometry. Moreover $\|r_{\vartheta_1} - r_{\vartheta_2}\|_{\mathcal{H}}$ is exactly the Hellinger distance between P_{ϑ_1} and P_{ϑ_2} up to the normalising convention, so DQM says that $\vartheta \mapsto P_\vartheta$ is differentiable *in Hellinger distance*. That is the metric one wants, because it interacts correctly with products of experiments and hence survives the localisation $h \mapsto h/\sqrt{n}$ that the whole asymptotic theory is built on.

Why the choice of μ *does not matter.* Suppose $d\mu = v\, d\mu'$ for a positive v. Then $p'_\vartheta = p_\vartheta v$ and $r'_\vartheta = r_\vartheta \sqrt{v}$, and the map $u \mapsto u\sqrt{v}$ is a linear isometry from $L^2(\mu)$ onto $L^2(\mu')$, since $\int |u\sqrt{v}|^2 d\mu' = \int u^2 v\, d\mu' = \int u^2 d\mu$. This isometry does not depend on ϑ, so it carries (2.35) to the corresponding statement for μ' and carries $\frac{1}{2}\dot\ell(h) r_\vartheta$ to $\frac{1}{2}\dot\ell(h) r'_\vartheta$, leaving the score $\dot\ell$ itself untouched. This is why [Sun et al.(2026)Sun, Lin, and Liu] can say that the dominating measure is not necessarily unique without any further comment.

Why the factor $\frac{1}{2}$*, and why the derivative is written as a multiple of* r_{ϑ_0}. The $\frac{1}{2}$ is the chain rule for the square root: where things are smooth enough to compute pointwise,

$$\frac{\partial}{\partial\vartheta}\sqrt{p_\vartheta} = \frac{1}{2\sqrt{p_\vartheta}}\frac{\partial p_\vartheta}{\partial\vartheta} = \frac{1}{2}\frac{\partial \log p_\vartheta}{\partial\vartheta}\sqrt{p_\vartheta}.$$

So the factor is there for exactly one reason: to make the object called the score the derivative of $\log p_\vartheta$ rather than of $\sqrt{p_\vartheta}$. As for the shape $\frac{1}{2}\dot\ell(h)\, r_{\vartheta_0}$, note that $u \mapsto u\, r_{\vartheta_0}$ is an isometry from $L^2(P_{\vartheta_0})$ into $L^2(\mu)$, because

$$\int (u\, r_{\vartheta_0})^2\, d\mu = \int u^2 p_{\vartheta_0}\, d\mu = \mathbb{E}\left[u(\boldsymbol{X})^2\right]. \tag{2.36}$$

Insisting that the derivative be written in the form $\frac{1}{2}\dot\ell(h)\, r_{\vartheta_0}$ rather than merely as some element of $L^2(\mu)$ is therefore a change of variables which factors the derivative through that isometry. It is what makes the score a *random variable*, square integrable under P_{ϑ_0} and independent of μ, instead of a μ-dependent function.

The model as a surface in the unit sphere

The picture to keep in mind is that a statistical model is a parametrised surface drawn on the unit sphere of a Hilbert space, and that the score is its tangent vector. Two standard facts are immediate consequences of that picture rather than of any computation with integrals.

First, *the score has mean zero because the derivative is tangent to the sphere*. Differentiating the identity $\|r_\vartheta\|^2 \equiv 1$ in the direction h gives

$$0 = 2\big\langle D\Phi_{\vartheta_0}(h),\, r_{\vartheta_0}\big\rangle_{\mathcal{H}} = \int \dot\ell_{\vartheta_0}(h)\, r^2_{\vartheta_0}\, d\mu = \int \dot\ell_{\vartheta_0}(h)\, p_{\vartheta_0}\, d\mu = \mathbb{E}\left[\dot\ell_{\vartheta_0}(h)\right]. \tag{2.37}$$

So the vanishing of the expected score is not an artifact of differentiating under the integral sign, and does not need the interchange to be justified: it is the constraint $\|r_\vartheta\| \equiv 1$ differentiated once. This is why the classical theory obtains it as a *consequence* of DQM rather than as a hypothesis (see [van der Vaart(1998)], Theorem 7.2). In our own development we verified it by direct computation in Lemma 13(2), which is a perfectly good proof, but (2.37) is the reason.

Second, *the Fisher information is the pullback of the* L^2 *inner product*. Indeed, by (2.36),

$$\big\langle D\Phi_{\vartheta_0}(h),\, D\Phi_{\vartheta_0}(h')\big\rangle_{\mathcal{H}} = \frac{1}{4}\int \dot\ell(h)\dot\ell(h')\, p_{\vartheta_0}\, d\mu = \frac{1}{4}\, G_{\vartheta_0}(h,h'). \tag{2.38}$$

So the Fisher form is four times the first fundamental form of the model regarded as a surface in $S(\mathcal{H})$. In particular

$$G_{\vartheta_0} \text{ is positive definite } \iff D\Phi_{\vartheta_0} \text{ is injective } \iff \Phi \text{ is an immersion at } \vartheta_0,$$

which is the geometric reading of Proposition 15: that proposition says exactly that our model, viewed through Φ, is an immersed surface in $S(\mathcal{H})$ rather than one that has been crushed in some direction.

Fréchet, not merely directional

It is worth being precise about which notion of differentiability is meant, because that is where the analytic work actually lies. Condition (2.35) is *Fréchet* differentiability: the remainder is $o(\|h\|)$ as $h \to \mathbf{0}$ in any manner whatsoever. It is strictly stronger than the directional, or Gateaux, statement

$$\lim_{t\to 0} \frac{1}{t}\|\Phi(\vartheta_0 + tv) - \Phi(\vartheta_0) - t\, D\Phi_{\vartheta_0}(v)\| = 0 \qquad \text{for each fixed } v,$$

which does not suffice for local asymptotic normality even when the directional derivative happens to be linear in v. The gap between the two is exactly a uniformity over directions. This is what the classical sufficient condition ([van der Vaart(1998)], Lemma 7.6) buys with its continuity hypothesis on the information matrix, and it is what Step 4 of the proof of Proposition 18 supplies by hand, by combining linearity in the direction with convergence at finitely many basis directions. A reader who wonders why that proof is not simply Step 3 now has the answer: Step 3 alone gives only the Gateaux statement.

What changes on a manifold

Almost nothing, conceptually. Comparing (2.8) with (2.34), the target Hilbert space is untouched; only the *source* has changed, from an open subset of $\mathbb{R}^k$ to the manifold M. Differentiability at a point of a map from a manifold into a Banach space is an entirely standard notion, and its derivative is a linear map $d\Phi_{[\boldsymbol{A},s]} : T_{[\boldsymbol{A},s]}M \to \mathcal{H}$.

The exponential map enters for one reason only: on a manifold there is no "$\vartheta_0 + h$", so to compare Φ at nearby points one must pass through some chart, and $\mathrm{Exp}_{[\boldsymbol{A},s]}$ is the canonical intrinsic choice of one. It is a convenience, not an ingredient. Concretely, if φ is any other C^1 chart centred at $[\boldsymbol{A}, s]$ whose differential at the origin is the identity of $T_{[\boldsymbol{A},s]}M$, then φ and $\mathrm{Exp}_{[\boldsymbol{A},s]}$ differ by a map tangent to the identity, which a remainder of size $o(\|h\|)$ cannot detect; so DQM through φ holds if and only if DQM through $\mathrm{Exp}_{[\boldsymbol{A},s]}$ holds, and the score operator is the same in both cases. For the same reason the condition does not depend on which norm is used to measure $\|h\|$, all norms on the finite dimensional space $T_{[\boldsymbol{A},s]}M$ being equivalent. The definition is thus considerably more robust than its notation suggests; what is genuinely intrinsic in it is the pair $\big(T_{[\boldsymbol{A},s]}M,\ d\Phi_{[\boldsymbol{A},s]}\big)$.

Remark 21. This robustness comes with a trap worth flagging, since M now carries *two* bilinear forms which are easily conflated. The quotient Euclidean metric g of (1.3) is the one that defines $\mathrm{Exp}_{[\boldsymbol{A},s]}$, $\mathrm{Exp}^{-1}_{[\boldsymbol{A},s]}$, the horizontal lift and the norm $\|h\|$ appearing in (2.8). The Fisher form $G_{[\boldsymbol{A},s]}$ of (2.14) is the statistical one, and it is the inverse $G^{-1}_{[\boldsymbol{A},s]}$ that will appear as an asymptotic covariance. These two are *not* proportional: g is a fixed Euclidean object, while G depends on $\boldsymbol{\Sigma}_0$ through $\boldsymbol{\Sigma}_0^{-1}\dot{\boldsymbol{\Sigma}}[\cdot]\boldsymbol{\Sigma}_0^{-1}$. Geodesics, logarithms and parallel transport are taken with respect to g; variances are read off G^{-1}. Nothing above is harmed by this, precisely because DQM is insensitive to the choice of g, but the distinction has to be maintained in any statement that mixes the two.

The present model: a factorization

For the probabilistic principal component analysis model the root density map admits a factorization that isolates where the geometry actually lives.

Proposition 22. Let $N : \mathrm{Sym}_{++}(p) \to L^2(\mathbb{R}^p, d\boldsymbol{x})$ send a positive definite matrix to the square root of the $N_p(\mathbf{0}, \cdot)$ density,

$$N(\boldsymbol{\Sigma}_1)(\boldsymbol{x}) := \frac{\exp\left(-\frac{1}{4}\boldsymbol{x}^T\boldsymbol{\Sigma}_1^{-1}\boldsymbol{x}\right)}{(2\pi)^{p/4}\det(\boldsymbol{\Sigma}_1)^{1/4}}.$$

Then the root density map $\Phi = r(\cdot\,;\cdot)$ of (2.3) factors as $\Phi = N \circ \boldsymbol{\Sigma}$, that is,

$$M \xrightarrow{\boldsymbol{\Sigma}} \mathrm{Sym}_{++}(p) \xrightarrow{N} S\left(L^2(\mathbb{R}^p, d\boldsymbol{x})\right),$$

with $\boldsymbol{\Sigma}$ as in (2.1). Moreover N is an injective immersion, and the metric it induces on $\mathrm{Sym}_{++}(p)$ is

$$4\big\langle dN_{\boldsymbol{\Sigma}_1}[\boldsymbol{\Delta}],\, dN_{\boldsymbol{\Sigma}_1}[\boldsymbol{\Delta}']\big\rangle_{L^2} = \frac{1}{2}\,\mathrm{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\,\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}'\right), \tag{2.39}$$

namely one half of the affine invariant metric of the positive definite cone. Consequently $G_{[\boldsymbol{A},s]}$ is the pullback along $\boldsymbol{\Sigma}$ of (2.39), and

$$G_{[\boldsymbol{A},s]} \text{ is positive definite } \iff \boldsymbol{\Sigma} : M \to \mathrm{Sym}_{++}(p) \text{ is an immersion at } [\boldsymbol{A}, s].$$

Proof. The factorization is the definition (2.2) of the model, which produces the density from the class $[\boldsymbol{A}, s]$ only through the matrix $\boldsymbol{\Sigma}[\boldsymbol{A}, s]$. Injectivity of N holds because distinct covariances give distinct centred Gaussian laws. For (2.39), apply (2.38) to the family $\{N_p(\mathbf{0}, \boldsymbol{\Sigma}_1)\}$ parametrised by $\boldsymbol{\Sigma}_1 \in \mathrm{Sym}_{++}(p)$ itself, whose score in the direction $\boldsymbol{\Delta}$ is $T_{\boldsymbol{\Delta}}$ in the notation (2.9) and whose information is therefore $\frac{1}{2}\,\mathrm{tr}\left(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\Delta}'\right)$ by Lemma 12. That form is positive definite by the computation of Lemma 13(3) applied at $\boldsymbol{\Sigma}_1$, so dN is injective and N is an immersion. The final two claims follow by the chain rule $d\Phi = dN \circ d\boldsymbol{\Sigma}$: comparing (2.14) with (2.39) identifies $G_{[\boldsymbol{A},s]}$ as $\boldsymbol{\Sigma}^*$ of the latter, and since dN is injective, $d\Phi_{[\boldsymbol{A},s]}$ is injective if and only if $d\boldsymbol{\Sigma}_{[\boldsymbol{A},s]}$ is. □

The point of Proposition 22 is that the second arrow is model independent: it is the same for any family of centred Gaussians, and it is an immersion for trivial reasons. All of the content specific to probabilistic principal component analysis sits in the first arrow $\boldsymbol{\Sigma}$, which is why the score (2.13), the Fisher form (2.14) and the chart covariance (2.4) are all expressed through $\dot{\boldsymbol{\Sigma}}$ and nothing else. It also puts Remark 14 in its proper light. On the total space Θ the map $\boldsymbol{\Sigma}$ is *not* an immersion: its differential kills the vertical space $V_{(\boldsymbol{A},s)}$, of dimension $\frac{q(q-1)}{2}$, as was established inside the proof of Proposition 1. That failure *is* the non-identifiability of the model. Passing to the quotient $M = \Theta/\sim$ is exactly the operation that repairs it, and the assertion that the repair is complete — that nothing beyond the vertical space is killed — is precisely Proposition 15. In the language of this section, then, that proposition says that $\boldsymbol{\Sigma}$ descends from a map which is merely constant on orbits to one which is an *immersion* of M, and hence that $\Phi = N \circ \boldsymbol{\Sigma}$ realises M as an immersed submanifold of the unit sphere of $L^2(\mathbb{R}^p, d\boldsymbol{x})$.

Two calibrating examples

Finally, two classical one dimensional examples show that the Fréchet in L^2 reading is doing real work, and in particular that DQM is neither implied by nor implies pointwise smoothness of the density.

The double exponential location family $p_\vartheta(x) = \frac{1}{2}e^{-|x-\vartheta|}$ on $\mathbb{R}$ is differentiable in quadratic mean at every ϑ, with score $\dot{\ell}_\vartheta(x) = \mathrm{sign}(x - \vartheta)$, even though $x \mapsto p_\vartheta(x)$ fails to be differentiable at $x = \vartheta$ for every ϑ. A single non-differentiability, being a μ-null event, is invisible to an L^2 norm.

Conversely, let $p_\vartheta = \vartheta^{-1}\mathbf{1}_{[0,\vartheta]}$ be the uniform family on $[0, \vartheta]$, $\vartheta > 0$. For $h > 0$ small,

$$\|r_{\vartheta+h} - r_\vartheta\|_{L^2}^2 = \int_0^\vartheta \left(\frac{1}{\sqrt{\vartheta+h}} - \frac{1}{\sqrt{\vartheta}}\right)^2 dx + \int_\vartheta^{\vartheta+h} \frac{dx}{\vartheta+h} = \frac{h^2}{4\vartheta^2} + O(h^3) + \frac{h}{\vartheta+h},$$

so that $\|r_{\vartheta+h} - r_\vartheta\|_{L^2} \sim \sqrt{h/\vartheta}$. The curve $\vartheta \mapsto r_\vartheta$ is therefore only Hölder of exponent $\frac{1}{2}$ in $\mathcal{H}$ and is nowhere differentiable, so no score operator can exist and the family is not DQM at any ϑ — notwithstanding that $p_\vartheta(x)$ is a perfectly smooth, indeed constant, function of ϑ for each fixed $x \neq \vartheta$. This is the analytic shadow of the familiar fact that the maximum likelihood estimator in this model converges at rate n^{-1} rather than $n^{-1/2}$, and it shows that the whole asymptotic theory really is keyed to the smoothness of a curve in a Hilbert space and not to the smoothness of a density.

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